APPLIED ECONOMETRICS PROJECT

Wage Determination, Regression Diagnostics and Time-Series Autocorrelation

Comprehensive report with available results, tables, figures, interpretations and source code

Report scope

This report assembles available outputs from the six-part project. It includes exported tables, figures found in the project folders, data-driven descriptions, and source code files available in the Colab working directory. Interpretations are limited to the exported values and do not treat descriptive associations as causal effects.

Contents

• Part 1 — Data preparation and exploratory analysis

• Part 2 — OLS and inference

• Part 3 — Functional forms and dummy variables

• Part 4 — Classical regression diagnostics

• Part 5 — Model specification and autocorrelation

• Part 6 — Consolidation and documentation

• Results tables and interpretations

• Figures

• Source code appendix

Results tables and interpretations

Table 1. Consolidated Table Inventory

Source: /content/final_project_outputs/consolidated_table_inventory.csv

part	source_file	rows	columns	consolidated_file	status
Part 1 — Data and EDA	outputs_part1/data_preparation_decisions.csv	7	2	final_project_outputs/tables/outputs_part1_data_preparation_decisions.csv	collected
Part 1 — Data and EDA	outputs_part1/dataset_summary.csv	2	9	final_project_outputs/tables/outputs_part1_dataset_summary.csv	collected
Part 1 — Data and EDA	outputs_part1/phillips_audit_summary.csv	9	2	final_project_outputs/tables/outputs_part1_phillips_audit_summary.csv	collected
Part 1 — Data and EDA	outputs_part1/phillips_derived_variable_checks.csv	4	2	final_project_outputs/tables/outputs_part1_phillips_derived_variable_checks.csv	collected
Part 1 — Data and EDA	outputs_part1/phillips_descriptive_statistics.csv	4	8	final_project_outputs/tables/outputs_part1_phillips_descriptive_statistics.csv	collected
Part 1 — Data and EDA	outputs_part1/phillips_numeric_integrity.csv	7	5	final_project_outputs/tables/outputs_part1_phillips_numeric_integrity.csv	collected
Part 1 — Data and EDA	outputs_part1/phillips_time_series_checks.csv	5	3	final_project_outputs/tables/outputs_part1_phillips_time_series_checks.csv	collected
Part 1 — Data and EDA	outputs_part1/phillips_working_data.csv	56	7	final_project_outputs/tables/outputs_part1_phillips_working_data.csv	collected
Part 1 — Data and EDA	outputs_part1/wage1_audit_summary.csv	9	2	final_project_outputs/tables/outputs_part1_wage1_audit_summary.csv	collected
Part 1 — Data and EDA	outputs_part1/wage1_correlation_matrix.csv	8	9	final_project_outputs/tables/outputs_part1_wage1_correlation_matrix.csv	collected
Part 1 — Data and EDA	outputs_part1/wage1_descriptive_statistics.csv	7	8	final_project_outputs/tables/outputs_part1_wage1_descriptive_statistics.csv	collected
Part 1 — Data and EDA	outputs_part1/wage1_group_characteristics.csv	4	5	final_project_outputs/tables/outputs_part1_wage1_group_characteristics.csv	collected
Part 1 — Data and EDA	outputs_part1/wage1_group_gender.csv	2	7	final_project_outputs/tables/outputs_part1_wage1_group_gender.csv	collected
Part 1 — Data and EDA	outputs_part1/wage1_group_gender_marital.csv	4	8	final_project_outputs/tables/outputs_part1_wage1_group_gender_marital.csv	collected

part	source_file	rows	columns	consolidated_file	status
Part 1 — Data and EDA	outputs_part1/wage1_group_marital_status.csv	2	7	final_project_outputs/tables/outputs_part1_wage1_group_marital_status.csv	collected
Part 1 — Data and EDA	outputs_part1/wage1_indicator_frequencies.csv	16	6	final_project_outputs/tables/outputs_part1_wage1_indicator_frequencies.csv	collected
Part 1 — Data and EDA	outputs_part1/wage1_numeric_integrity.csv	24	5	final_project_outputs/tables/outputs_part1_wage1_numeric_integrity.csv	collected
Part 1 — Data and EDA	outputs_part1/wage1_plausibility_flags.csv	9	2	final_project_outputs/tables/outputs_part1_wage1_plausibility_flags.csv	collected
Part 1 — Data and EDA	outputs_part1/wage1_variable_presence.csv	7	3	final_project_outputs/tables/outputs_part1_wage1_variable_presence.csv	collected
Part 1 — Data and EDA	outputs_part1/wage1_working_data.csv	526	25	final_project_outputs/tables/outputs_part1_wage1_working_data.csv	collected
Part 1 — Data and EDA	outputs_part1/wage_profile.csv	9	2	final_project_outputs/tables/outputs_part1_wage_profile.csv	collected
Part 2 — OLS and inference	outputs_part2/tables/individual_t_tests.csv	9	9	final_project_outputs/tables/outputs_part2_individual_t_tests.csv	collected
Part 2 — OLS and inference	outputs_part2/tables/joint_f_test_experience_tenure.csv	1	7	final_project_outputs/tables/outputs_part2_joint_f_test_experience_tenure.csv	collected
Part 2 — OLS and inference	outputs_part2/tables/model_sample_descriptive_statistics.csv	7	7	final_project_outputs/tables/outputs_part2_model_sample_descriptive_statistics.csv	collected
Part 2 — OLS and inference	outputs_part2/tables/multiple_model_interpretation.csv	6	5	final_project_outputs/tables/outputs_part2_multiple_model_interpretation.csv	collected
Part 2 — OLS and inference	outputs_part2/tables/ols_coefficient_comparison.csv	9	8	final_project_outputs/tables/outputs_part2_ols_coefficient_comparison.csv	collected
Part 2 — OLS and inference	outputs_part2/tables/ols_model_fit_comparison.csv	2	8	final_project_outputs/tables/outputs_part2_ols_model_fit_comparison.csv	collected
Part 2 — OLS and inference	outputs_part2/tables/overall_f_tests.csv	2	7	final_project_outputs/tables/outputs_part2_overall_f_tests.csv	collected
Part 2 — OLS and inference	outputs_part2/tables/sample_summary.csv	1	4	final_project_outputs/tables/outputs_part2_sample_summary.csv	collected
Part 3 — Functional forms and dummy variables	outputs_part3/tables/education_indicator_interaction.csv	8	8	final_project_outputs/tables/outputs_part3_education_indicator_interaction.csv	collected

part	source_file	rows	columns	consolidated_file	status
Part 3 — Functional forms and dummy variables	outputs_part3/tables/functional_form_coefficients.csv	14	8	final_project_outputs/tables/outputs_part3_functional_form_coefficients.csv	collected
Part 3 — Functional forms and dummy variables	outputs_part3/tables/functional_form_model_summary.csv	2	6	final_project_outputs/tables/outputs_part3_functional_form_model_summary.csv	collected
Part 3 — Functional forms and dummy variables	outputs_part3/tables/joint_dummy_f_test.csv	1	7	final_project_outputs/tables/outputs_part3_joint_dummy_f_test.csv	collected
Part 3 — Functional forms and dummy variables	outputs_part3/tables/log_wage_coefficient_interpretation.csv	6	4	final_project_outputs/tables/outputs_part3_log_wage_coefficient_interpretation.csv	collected
Part 3 — Functional forms and dummy variables	outputs_part3/tables/mwd_functional_form_test.csv	1	2	final_project_outputs/tables/outputs_part3_mwd_functional_form_test.csv	collected
Part 4 — Regression diagnostics	outputs_part4/tables/conventional_vs_robust_standard_errors.csv	7	6	final_project_outputs/tables/outputs_part4_conventional_vs_robust_standard_errors.csv	collected
Part 4 — Regression diagnostics	outputs_part4/tables/diagnostic_summary.csv	1	7	final_project_outputs/tables/outputs_part4_diagnostic_summary.csv	collected
Part 4 — Regression diagnostics	outputs_part4/tables/heteroskedasticity_tests.csv	2	6	final_project_outputs/tables/outputs_part4_heteroskedasticity_tests.csv	collected
Part 4 — Regression diagnostics	outputs_part4/tables/jarque_bera_normality_test.csv	1	5	final_project_outputs/tables/outputs_part4_jarque_bera_normality_test.csv	collected
Part 4 — Regression diagnostics	outputs_part4/tables/predictor_correlation_matrix.csv	6	7	final_project_outputs/tables/outputs_part4_predictor_correlation_matrix.csv	collected
Part 4 — Regression diagnostics	outputs_part4/tables/variance_inflation_factors.csv	6	2	final_project_outputs/tables/outputs_part4_variance_inflation_factors.csv	collected
Part 5 — Specification and autocorrelation	outputs_part5/tables/autocorrelation_tests.csv	2	5	final_project_outputs/tables/outputs_part5_autocorrelation_tests.csv	collected
Part 5 — Specification and autocorrelation	outputs_part5/tables/breusch_godfrey_f_test.csv	1	4	final_project_outputs/tables/outputs_part5_breusch_godfrey_f_test.csv	collected
Part 5 — Specification and autocorrelation	outputs_part5/tables/joint_f_test_selected_regressors.csv	1	7	final_project_outputs/tables/outputs_part5_joint_f_test_selected_regressors.csv	collected
Part 5 — Specification and autocorrelation	outputs_part5/tables/phillips_time_series_data_notes.csv	1	6	final_project_outputs/tables/outputs_part5_phillips_time_series_data_notes.csv	collected

part	source_file	rows	columns	consolidated_file	status
Part 5 — Specification and autocorrelation	outputs_part5/tables/ramsey_reset_test.csv	1	5	final_project_outputs/tables/outputs_part5_ramsey_reset_test.csv	collected

Interpretation

This table contains 46 rows and 6 columns.

rows: 46 non-missing numeric entries; minimum 1, maximum 526, mean 17.8.

columns: 46 non-missing numeric entries; minimum 2, maximum 25, mean 6.022.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 2. Outputs Part1 Data Preparation Decisions

Source: /content/final_project_outputs/tables/outputs_part1_data_preparation_decisions.csv

Item	Decision
Row deletion (both datasets)	No observations are removed. Unusual values are flagged, not treated as errors.
Outliers in wage	Retained. High wages are plausible in survey data; any robustness treatment is deferred and would be justified explicitly.
Missing values - wage1	0 missing cells found. No imputation needed.
Missing values - phillips	4 missing cells found, all in lagged/differenced variables for the first year (by construction). Not imputed; Part 5 will use the lags it actually needs.
Duplicate rows	wage1 has 0 and phillips has 0 exact duplicate rows. Not deleted (no ID variable to establish true duplication).
New variable: log_wage	Created as natural log of wage for strictly positive wages. Observations with wage <= 0 would receive NaN (currently 0). Original 'wage' and 'lwage' are preserved.
Existing transformed variables	lwage, expersq and tenursq (supplied by the dataset) are kept unchanged; they are verified against the originals in Cell 9.

Interpretation

This table contains 7 rows and 2 columns.

This is a categorical/text summary table; interpret its entries using the row and column labels.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 3. Outputs Part1 Dataset Summary

Source: /content/final_project_outputs/tables/outputs_part1_dataset_summary.csv

Dataset	Observations	Variables	Numeric variables	Missing cells	Duplicate rows	Infinite values	Time information
wage1	526	24	24	0	0	0	none (cross-section)
phillips	56	7	7	4	0	0	annual, 1948-2003

Display limited to first 100 rows and first 8 columns; complete data remain in the CSV/workbook.

Interpretation

This table contains 2 rows and 9 columns.

wage1: 526 observations and 24 variables.

phillips: 56 observations and 7 variables.

These counts describe the datasets used for the project's cross-sectional wage analysis and supplementary time-series analysis.

Table 4. Outputs Part1 Phillips Audit Summary

Source: /content/final_project_outputs/tables/outputs_part1_phillips_audit_summary.csv

metric	value
n_obs	56
n_vars	7
missing_cells	4
missing_by_col	{'inf_1': 1, 'unem_1': 1, 'cinf': 1, 'cunem': 1}
n_duplicate_rows	0
n_inf	0
non_numeric_cols	[]
numeric_cols	['year', 'unem', 'inf', 'inf_1', 'unem_1', 'cinf', 'cunem']
flag_counts	{'unem < 0 or unem > 100': 0, '|inf| > 20 (heuristic screen)': 0}

Interpretation

This table contains 9 rows and 2 columns.

This is a categorical/text summary table; interpret its entries using the row and column labels.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 5. Outputs Part1 Phillips Derived Variable Checks

Source: /content/final_project_outputs/tables/outputs_part1_phillips_derived_variable_checks.csv

Unnamed: 0	max absolute difference
inf_1 vs inf shifted by one year	0
unem_1 vs unem shifted by one year	0
cinf vs inf - inf_1	2.384e-07
cunem vs unem - unem_1	1.776e-15

Interpretation

This table contains 4 rows and 2 columns.

max absolute difference: 4 non-missing numeric entries; minimum 0, maximum 2.384e-07, mean 5.96e-08.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 6. Outputs Part1 Phillips Descriptive Statistics

Source: /content/final_project_outputs/tables/outputs_part1_phillips_descriptive_statistics.csv

Unnamed: 0	documented meaning	count	mean	median	std	min	max
unem	civilian unemployment rate, %	56	5.6339	5.55	1.5088	2.9	9.7
inf	percentage change in CPI	56	3.8839	3.05	3.0404	-1.2	13.5
cunem	unem - unem_1	55	0.04	-0.2	1.0642	-2.1	2.9
cinf	inf - inf_1	55	-0.1055	0.2	2.414	-9.3	6.6

Interpretation

This table contains 4 rows and 8 columns.

count: 4 non-missing numeric entries; minimum 55, maximum 56, mean 55.5.

mean: 4 non-missing numeric entries; minimum -0.1055, maximum 5.634, mean 2.363.

median: 4 non-missing numeric entries; minimum -0.2, maximum 5.55, mean 2.15.

std: 4 non-missing numeric entries; minimum 1.064, maximum 3.04, mean 2.007.

min: 4 non-missing numeric entries; minimum -9.3, maximum 2.9, mean -2.425.

max: 4 non-missing numeric entries; minimum 2.9, maximum 13.5, mean 8.175.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 7. Outputs Part1 Phillips Numeric Integrity

Source: /content/final_project_outputs/tables/outputs_part1_phillips_numeric_integrity.csv

variable	numeric_dtype	entries_not_convertible_to_number	infinite_values	missing_values
year	True	0	0	0
unem	True	0	0	0
inf	True	0	0	0
inf_1	True	0	0	1
unem_1	True	0	0	1
cinf	True	0	0	1
cunem	True	0	0	1

Interpretation

This table contains 7 rows and 5 columns.

entries_not_convertible_to_number: 7 non-missing numeric entries; minimum 0, maximum 0, mean 0.

infinite_values: 7 non-missing numeric entries; minimum 0, maximum 0, mean 0.

missing_values: 7 non-missing numeric entries; minimum 0, maximum 1, mean 0.5714.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 8. Outputs Part1 Phillips Time Series Checks

Source: /content/final_project_outputs/tables/outputs_part1_phillips_time_series_checks.csv

Unnamed: 0	check	passed
0	year_whole_numbers	True
1	year_sorted	True
2	year_unique	True
3	year_consecutive	True
4	derived_variables_consistent	True

Interpretation

This table contains 5 rows and 3 columns.

Unnamed: 0: 5 non-missing numeric entries; minimum 0, maximum 4, mean 2.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 9. Outputs Part1 Phillips Working Data

Source: /content/final_project_outputs/tables/outputs_part1_phillips_working_data.csv

year	unem	inf	inf_1	unem_1	cinf	cunem
1948	3.8	8.1				
1949	5.9	-1.2	8.1	3.8	-9.3	2.1

year	unem	inf	inf_1	unem_1	cinf	cunem
1950	5.3	1.3	-1.2	5.9	2.5	-0.6
1951	3.3	7.9	1.3	5.3	6.6	-2
1952	3	1.9	7.9	3.3	-6	-0.3
1953	2.9	0.8	1.9	3	-1.1	-0.1
1954	5.5	0.7	0.8	2.9	-0.1	2.6
1955	4.4	-0.4	0.7	5.5	-1.1	-1.1
1956	4.1	1.5	-0.4	4.4	1.9	-0.3
1957	4.3	3.3	1.5	4.1	1.8	0.2
1958	6.8	2.8	3.3	4.3	-0.5	2.5
1959	5.5	0.7	2.8	6.8	-2.1	-1.3
1960	5.5	1.7	0.7	5.5	1	0
1961	6.7	1	1.7	5.5	-0.7	1.2
1962	5.5	1	1	6.7	0	-1.2
1963	5.7	1.3	1	5.5	0.3	0.2
1964	5.2	1.3	1.3	5.7	0	-0.5
1965	4.5	1.6	1.3	5.2	0.3	-0.7
1966	3.8	2.9	1.6	4.5	1.3	-0.7
1967	3.8	3.1	2.9	3.8	0.2	0
1968	3.6	4.2	3.1	3.8	1.1	-0.2
1969	3.5	5.5	4.2	3.6	1.3	-0.1
1970	4.9	5.7	5.5	3.5	0.2	1.4
1971	5.9	4.4	5.7	4.9	-1.3	1
1972	5.6	3.2	4.4	5.9	-1.2	-0.3
1973	4.9	6.2	3.2	5.6	3	-0.7
1974	5.6	11	6.2	4.9	4.8	0.7
1975	8.5	9.1	11	5.6	-1.9	2.9
1976	7.7	5.8	9.1	8.5	-3.3	-0.8
1977	7.1	6.5	5.8	7.7	0.7	-0.6
1978	6.1	7.6	6.5	7.1	1.1	-1
1979	5.8	11.3	7.6	6.1	3.7	-0.3
1980	7.1	13.5	11.3	5.8	2.2	1.3
1981	7.6	10.3	13.5	7.1	-3.2	0.5
1982	9.7	6.2	10.3	7.6	-4.1	2.1
1983	9.6	3.2	6.2	9.7	-3	-0.1
1984	7.5	4.3	3.2	9.6	1.1	-2.1
1985	7.2	3.6	4.3	7.5	-0.7	-0.3
1986	7	1.9	3.6	7.2	-1.7	-0.2
1987	6.2	3.6	1.9	7	1.7	-0.8
1988	5.5	4.1	3.6	6.2	0.5	-0.7
1989	5.3	4.8	4.1	5.5	0.7	-0.2
1990	5.6	5.4	4.8	5.3	0.6	0.3
1991	6.8	4.2	5.4	5.6	-1.2	1.2
1992	7.5	3	4.2	6.8	-1.2	0.7
1993	6.9	3	3	7.5	0	-0.6

year	unem	inf	inf_1	unem_1	cinf	cunem
1994	6.1	2.6	3	6.9	-0.4	-0.8
1995	5.6	2.8	2.6	6.1	0.2	-0.5
1996	5.4	3	2.8	5.6	0.2	-0.2
1997	4.9	2.3	3	5.4	-0.7	-0.5
1998	4.5	1.6	2.3	4.9	-0.7	-0.4
1999	4.2	2.2	1.6	4.5	0.6	-0.3
2000	4	3.4	2.2	4.2	1.2	-0.2
2001	4.8	2.8	3.4	4	-0.6	0.8
2002	5.8	1.6	2.8	4.8	-1.2	1
2003	6	2.3	1.6	5.8	0.7	0.2

Interpretation

This table contains 56 rows and 7 columns.

year: 56 non-missing numeric entries; minimum 1948, maximum 2003, mean 1976.

unem: 56 non-missing numeric entries; minimum 2.9, maximum 9.7, mean 5.634.

inf: 56 non-missing numeric entries; minimum -1.2, maximum 13.5, mean 3.884.

inf_1: 55 non-missing numeric entries; minimum -1.2, maximum 13.5, mean 3.913.

unem_1: 55 non-missing numeric entries; minimum 2.9, maximum 9.7, mean 5.627.

cinf: 55 non-missing numeric entries; minimum -9.3, maximum 6.6, mean -0.1055.

cunem: 55 non-missing numeric entries; minimum -2.1, maximum 2.9, mean 0.04.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 10. Outputs Part1 Wage1 Audit Summary

Source: /content/final_project_outputs/tables/outputs_part1_wage1_audit_summary.csv

metric	value
n_obs	526
n_vars	24
missing_cells	0
missing_by_col	{}
n_duplicate_rows	0
n_inf	0
non_numeric_cols	[]
numeric_cols	['wage', 'educ', 'exper', 'tenure', 'nonwhite', 'female', 'married', 'numdep', 'smsa', 'northcen', 'south', 'west', 'construc', 'ndurman', 'trcommpu', 'trade', 'services', 'profserv', 'profocc', 'clerocc', 'servocc', 'lwage', 'expersq', 'tenursq']
flag_counts	{'wage <= 0': 0, 'educ < 0 or educ > 20': 0, 'exper < 0 or exper > 60': 0, 'tenure < 0 or tenure > 50': 0, 'tenure > exper (tenure exceeds potential experience)': 16, 'indicator value outside {0,1}': 0, 'lwage differs from log(wage) (tolerance 1e-6)': 0, 'expersq differs from exper^2': 0, 'tenursq differs from tenure^2': 0}

Interpretation

This table contains 9 rows and 2 columns.

This is a categorical/text summary table; interpret its entries using the row and column labels.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 11. Outputs Part1 Wage1 Correlation Matrix

Source: /content/final_project_outputs/tables/outputs_part1_wage1_correlation_matrix.csv

Unnamed: 0	wage	log_wage	educ	exper	tenure	female	married
wage	1	0.9371	0.4059	0.1129	0.3469	-0.3401	0.2288
log_wage	0.9371	1	0.4311	0.1114	0.3255	-0.3737	0.2707
educ	0.4059	0.4311	1	-0.2995	-0.0562	-0.085	0.0689
exper	0.1129	0.1114	-0.2995	1	0.4993	-0.0416	0.317
tenure	0.3469	0.3255	-0.0562	0.4993	1	-0.1979	0.2399
female	-0.3401	-0.3737	-0.085	-0.0416	-0.1979	1	-0.1661
married	0.2288	0.2707	0.0689	0.317	0.2399	-0.1661	1
nonwhite	-0.0385	-0.0389	-0.0847	0.0144	0.0116	-0.0109	-0.0623

Display limited to first 100 rows and first 8 columns; complete data remain in the CSV/workbook.

Interpretation

This table contains 8 rows and 9 columns.

wage: 8 non-missing numeric entries; minimum -0.3401, maximum 1, mean 0.3316.

log_wage: 8 non-missing numeric entries; minimum -0.3737, maximum 1, mean 0.3329.

educ: 8 non-missing numeric entries; minimum -0.2995, maximum 1, mean 0.1726.

exper: 8 non-missing numeric entries; minimum -0.2995, maximum 1, mean 0.2142.

tenure: 8 non-missing numeric entries; minimum -0.1979, maximum 1, mean 0.2711.

female: 8 non-missing numeric entries; minimum -0.3737, maximum 1, mean -0.02692.

married: 8 non-missing numeric entries; minimum -0.1661, maximum 1, mean 0.2371.

nonwhite: 8 non-missing numeric entries; minimum -0.08465, maximum 1, mean 0.09884.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 12. Outputs Part1 Wage1 Descriptive Statistics

Source: /content/final_project_outputs/tables/outputs_part1_wage1_descriptive_statistics.csv

Unnamed: 0	documented meaning	count	mean	median	std	min	max
wage	average hourly earnings	526	5.8961	4.65	3.6931	0.53	24.98
educ	years of education	526	12.5627	12	2.769	0	18
exper	years potential experience	526	17.0171	13.5	13.5722	1	51
tenure	years with current employer	526	5.1046	2	7.2245	0	44
female	=1 if female	526	0.4791	0	0.5	0	1
married	=1 if married	526	0.6084	1	0.4886	0	1
nonwhite	=1 if nonwhite	526	0.1027	0	0.3038	0	1

Interpretation

This table contains 7 rows and 8 columns.

count: 7 non-missing numeric entries; minimum 526, maximum 526, mean 526.

mean: 7 non-missing numeric entries; minimum 0.1027, maximum 17.02, mean 5.967.

median: 7 non-missing numeric entries; minimum 0, maximum 13.5, mean 4.736.

std: 7 non-missing numeric entries; minimum 0.3038, maximum 13.57, mean 4.079.

min: 7 non-missing numeric entries; minimum 0, maximum 1, mean 0.2186.

max: 7 non-missing numeric entries; minimum 1, maximum 51, mean 20.14.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 13. Outputs Part1 Wage1 Group Characteristics

Source: /content/final_project_outputs/tables/outputs_part1_wage1_group_characteristics.csv

female	married	educ	exper	tenure
0	0	12.5465	10.7093	3.2558
0	1	12.8989	20.6915	7.9468
1	0	12.1667	12.3417	2.725
1	1	12.4545	20.1439	4.4242

Interpretation

This table contains 4 rows and 5 columns.

female: 4 non-missing numeric entries; minimum 0, maximum 1, mean 0.5.

married: 4 non-missing numeric entries; minimum 0, maximum 1, mean 0.5.

educ: 4 non-missing numeric entries; minimum 12.17, maximum 12.9, mean 12.52.

exper: 4 non-missing numeric entries; minimum 10.71, maximum 20.69, mean 15.97.

tenure: 4 non-missing numeric entries; minimum 2.725, maximum 7.947, mean 4.588.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 14. Outputs Part1 Wage1 Group Gender

Source: /content/final_project_outputs/tables/outputs_part1_wage1_group_gender.csv

female	count	mean	median	std	min	max
0	274	7.0995	6	4.1609	1.5	24.98
1	252	4.5877	3.75	2.5294	0.53	21.63

Interpretation

This table contains 2 rows and 7 columns.

female: 2 non-missing numeric entries; minimum 0, maximum 1, mean 0.5.

count: 2 non-missing numeric entries; minimum 252, maximum 274, mean 263.

mean: 2 non-missing numeric entries; minimum 4.588, maximum 7.099, mean 5.844.

median: 2 non-missing numeric entries; minimum 3.75, maximum 6, mean 4.875.

std: 2 non-missing numeric entries; minimum 2.529, maximum 4.161, mean 3.345.

min: 2 non-missing numeric entries; minimum 0.53, maximum 1.5, mean 1.015.

max: 2 non-missing numeric entries; minimum 21.63, maximum 24.98, mean 23.3.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 15. Outputs Part1 Wage1 Group Gender Marital

Source: /content/final_project_outputs/tables/outputs_part1_wage1_group_gender_marital.csv

female	married	count	mean	median	std	min	max
0	0	86	5.168	4.41	2.7228	1.5	12.5
0	1	188	7.983	6.5	4.405	2.23	24.98
1	0	120	4.6116	3.5	2.9956	0.53	21.63

female	married	count	mean	median	std	min	max
1	1	132	4.5659	4	2.0263	1.43	15

Interpretation

This table contains 4 rows and 8 columns.

female: 4 non-missing numeric entries; minimum 0, maximum 1, mean 0.5.

married: 4 non-missing numeric entries; minimum 0, maximum 1, mean 0.5.

count: 4 non-missing numeric entries; minimum 86, maximum 188, mean 131.5.

mean: 4 non-missing numeric entries; minimum 4.566, maximum 7.983, mean 5.582.

median: 4 non-missing numeric entries; minimum 3.5, maximum 6.5, mean 4.603.

std: 4 non-missing numeric entries; minimum 2.026, maximum 4.405, mean 3.037.

min: 4 non-missing numeric entries; minimum 0.53, maximum 2.23, mean 1.422.

max: 4 non-missing numeric entries; minimum 12.5, maximum 24.98, mean 18.53.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 16. Outputs Part1 Wage1 Group Marital Status

Source: /content/final_project_outputs/tables/outputs_part1_wage1_group_marital_status.csv

married	count	mean	median	std	min	max
0	206	4.8439	3.75	2.8911	0.53	21.63
1	320	6.5735	5.49	3.9874	1.43	24.98

Interpretation

This table contains 2 rows and 7 columns.

married: 2 non-missing numeric entries; minimum 0, maximum 1, mean 0.5.

count: 2 non-missing numeric entries; minimum 206, maximum 320, mean 263.

mean: 2 non-missing numeric entries; minimum 4.844, maximum 6.573, mean 5.709.

median: 2 non-missing numeric entries; minimum 3.75, maximum 5.49, mean 4.62.

std: 2 non-missing numeric entries; minimum 2.891, maximum 3.987, mean 3.439.

min: 2 non-missing numeric entries; minimum 0.53, maximum 1.43, mean 0.98.

max: 2 non-missing numeric entries; minimum 21.63, maximum 24.98, mean 23.3.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 17. Outputs Part1 Wage1 Indicator Frequencies

Source: /content/final_project_outputs/tables/outputs_part1_wage1_indicator_frequencies.csv

Unnamed: 0	variable	documented	count_0	count_1	share_1_%
0	nonwhite	=1 if nonwhite	472	54	10.2662
1	female	=1 if female	274	252	47.9087
2	married	=1 if married	206	320	60.8365
3	smsa	=1 if live in SMSA	146	380	72.2433
4	northcen	=1 if live in north central U.S	394	132	25.0951
5	south	=1 if live in southern region	339	187	35.5513
6	west	=1 if live in western region	437	89	16.9202

Unnamed: 0	variable	documented	count_0	count_1	share_1_%
7	construc	=1 if work in construc. indus.	502	24	4.5627
8	ndurman	=1 if in nondur. manuf. indus.	466	60	11.4068
9	trcommpu	=1 if in trans, commun, pub ut	503	23	4.3726
10	trade	=1 if in wholesale or retail	375	151	28.7072
11	services	=1 if in services indus.	473	53	10.076
12	profserv	=1 if in prof. serv. indus.	390	136	25.8555
13	profocc	=1 if in profess. occupation	333	193	36.692
14	clerocc	=1 if in clerical occupation	438	88	16.73
15	servocc	=1 if in service occupation	452	74	14.0684

Interpretation

This table contains 16 rows and 6 columns.

Unnamed: 0: 16 non-missing numeric entries; minimum 0, maximum 15, mean 7.5.

count_0: 16 non-missing numeric entries; minimum 146, maximum 503, mean 387.5.

count_1: 16 non-missing numeric entries; minimum 23, maximum 380, mean 138.5.

share_1_%: 16 non-missing numeric entries; minimum 4.373, maximum 72.24, mean 26.33.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 18. Outputs Part1 Wage1 Numeric Integrity

Source: /content/final_project_outputs/tables/outputs_part1_wage1_numeric_integrity.csv

variable	numeric_dtype	entries_not_convertible_to_number	infinite_values	missing_values
wage	True	0	0	0
educ	True	0	0	0
exper	True	0	0	0
tenure	True	0	0	0
nonwhite	True	0	0	0
female	True	0	0	0
married	True	0	0	0
numdep	True	0	0	0
smsa	True	0	0	0
northcen	True	0	0	0
south	True	0	0	0
west	True	0	0	0
construc	True	0	0	0
ndurman	True	0	0	0
trcommpu	True	0	0	0
trade	True	0	0	0
services	True	0	0	0

variable	numeric_dtype	entries_not_convertible_to_number	infinite_values	missing_values
profserv	True	0	0	0
profocc	True	0	0	0
clerocc	True	0	0	0
servocc	True	0	0	0
lwage	True	0	0	0
expersq	True	0	0	0
tenursq	True	0	0	0

Interpretation

This table contains 24 rows and 5 columns.

entries_not_convertible_to_number: 24 non-missing numeric entries; minimum 0, maximum 0, mean 0.

infinite_values: 24 non-missing numeric entries; minimum 0, maximum 0, mean 0.

missing_values: 24 non-missing numeric entries; minimum 0, maximum 0, mean 0.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 19. Outputs Part1 Wage1 Plausibility Flags

Source: /content/final_project_outputs/tables/outputs_part1_wage1_plausibility_flags.csv

Unnamed: 0	rows_flagged
wage <= 0	0
educ < 0 or educ > 20	0
exper < 0 or exper > 60	0
tenure < 0 or tenure > 50	0
tenure > exper (tenure exceeds potential experience)	16
indicator value outside {0,1}	0
lwage differs from log(wage) (tolerance 1e-6)	0
expersq differs from exper^2	0
tenursq differs from tenure^2	0

Interpretation

This table contains 9 rows and 2 columns.

rows_flagged: 9 non-missing numeric entries; minimum 0, maximum 16, mean 1.778.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 20. Outputs Part1 Wage1 Variable Presence

Source: /content/final_project_outputs/tables/outputs_part1_wage1_variable_presence.csv

Unnamed: 0	present	documented_meaning
wage	True	average hourly earnings
educ	True	years of education
exper	True	years potential experience
tenure	True	years with current employer
female	True	=1 if female
married	True	=1 if married
nonwhite	True	=1 if nonwhite

Interpretation

This table contains 7 rows and 3 columns.

This is a categorical/text summary table; interpret its entries using the row and column labels.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 21. Outputs Part1 Wage1 Working Data

Source: /content/final_project_outputs/tables/outputs_part1_wage1_working_data.csv

wage	educ	exper	tenure	nonwhite	female	married	numdep
3.1	11	2	0	0	1	0	2
3.24	12	22	2	0	1	1	3
3	11	2	0	0	0	0	2
6	8	44	28	0	0	1	0
5.3	12	7	2	0	0	1	1
8.75	16	9	8	0	0	1	0
11.25	18	15	7	0	0	0	0
5	12	5	3	0	1	0	0
3.6	12	26	4	0	1	0	2
18.18	17	22	21	0	0	1	0
6.25	16	8	2	0	1	0	0
8.13	13	3	0	0	1	0	0
8.77	12	15	0	0	0	1	2
5.5	12	18	3	0	0	0	0
22.2	12	31	15	0	0	1	1
17.33	16	14	0	0	0	1	1
7.5	12	10	0	0	1	1	0
10.63	13	16	10	0	1	0	0
3.6	12	13	0	0	1	1	3
4.5	12	36	6	0	1	1	0
6.88	12	11	4	0	1	0	0
8.48	12	29	13	0	0	1	3
6.33	16	9	9	0	1	0	0
0.53	12	3	1	0	1	0	0
6	11	37	8	1	1	0	0
9.56	16	3	3	1	0	1	1
7.78	16	11	10	0	0	1	1
12.5	16	31	0	0	0	1	0
12.5	15	30	0	0	0	1	2
3.25	8	9	1	0	1	1	2
13	14	23	5	0	0	1	2
4.5	14	2	5	0	1	0	2
9.68	13	16	16	0	1	0	0
5	12	7	3	0	1	0	0
4.68	12	3	0	0	1	0	0
4.27	16	22	4	1	1	1	1

wage	educ	exper	tenure	nonwhite	female	married	numdep
6.15	12	15	6	1	1	1	1
3.51	4	39	15	0	0	1	5
3	14	3	3	0	1	1	0
6.25	12	11	0	0	1	0	0
7.81	12	3	0	0	1	1	0
10	12	20	5	1	1	1	4
4.5	14	16	0	0	1	1	1
4	11	45	12	0	1	1	0
6.38	13	11	4	0	1	0	0
13.7	15	20	13	0	0	1	2
1.67	10	1	0	0	0	0	1
2.93	12	36	2	0	1	1	1
3.65	14	9	2	0	0	0	0
2.9	12	15	1	0	1	1	2
1.63	12	18	0	0	1	0	2
8.6	16	3	2	0	1	0	0
5	12	15	5	0	0	1	1
6	12	7	7	0	0	1	0
2.5	12	2	0	0	0	0	2
3.25	15	3	0	0	0	0	1
3.4	16	1	1	0	1	0	1
10	8	13	0	1	0	0	0
21.63	18	8	8	0	1	0	0
4.38	16	7	0	0	0	0	0
11.71	13	40	20	0	1	0	0
12.39	14	42	5	0	0	0	0
6.25	10	36	8	0	0	0	0
3.71	10	13	0	1	1	0	4
7.78	14	9	3	0	0	0	0
19.98	14	26	23	0	0	1	2
6.25	16	7	4	1	1	1	3
10	12	25	3	0	0	1	3
5.71	16	10	5	0	0	1	1
2	12	3	2	0	1	0	4
5.71	16	3	0	0	1	0	0
13.08	17	17	2	1	0	1	3
4.91	12	17	8	0	0	1	2
2.91	12	20	34	0	1	1	2
3.75	12	7	0	0	1	1	0
11.9	13	24	19	0	0	1	2
4	12	28	0	0	1	1	1
3.1	12	2	1	0	1	0	1
8.45	12	19	13	0	0	1	4
7.14	18	13	0	0	0	1	2

wage	educ	exper	tenure	nonwhite	female	married	numdep
4.5	9	22	5	0	0	1	4
4.65	16	3	1	0	0	0	0
2.9	10	4	0	0	1	0	1
6.67	12	7	5	0	0	0	0
3.5	12	6	2	1	1	0	1
3.26	12	13	3	0	1	1	1
3.25	12	14	0	0	1	1	1
8	12	14	4	0	1	0	1
9.85	8	40	24	1	0	1	2
7.5	12	11	7	0	0	1	1
5.91	12	14	6	0	0	1	2
11.76	14	40	39	0	0	1	0
3	12	1	0	0	1	0	2
4.81	12	2	0	0	1	0	0
6.5	12	4	1	0	1	0	0
4	9	19	1	0	1	1	1
3.5	13	1	0	0	0	0	1
13.16	12	34	22	0	0	1	0
4.25	14	5	2	0	0	1	2
3.5	12	3	0	0	0	0	0

Display limited to first 100 rows and first 8 columns; complete data remain in the CSV/workbook.

Interpretation

This table contains 526 rows and 25 columns.

wage: 526 non-missing numeric entries; minimum 0.53, maximum 24.98, mean 5.896.

educ: 526 non-missing numeric entries; minimum 0, maximum 18, mean 12.56.

exper: 526 non-missing numeric entries; minimum 1, maximum 51, mean 17.02.

tenure: 526 non-missing numeric entries; minimum 0, maximum 44, mean 5.105.

nonwhite: 526 non-missing numeric entries; minimum 0, maximum 1, mean 0.1027.

female: 526 non-missing numeric entries; minimum 0, maximum 1, mean 0.4791.

married: 526 non-missing numeric entries; minimum 0, maximum 1, mean 0.6084.

numdep: 526 non-missing numeric entries; minimum 0, maximum 6, mean 1.044.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 22. Outputs Part1 Wage Profile

Source: /content/final_project_outputs/tables/outputs_part1_wage_profile.csv

Unnamed: 0	wage
count	526
min	0.53
1st percentile	1.69
median	4.65
99th percentile	19.995
max	24.98
values <= 0	0

Unnamed: 0	wage
values missing	0
values above Tukey upper fence (Q3 + 1.5 IQR)	36

Interpretation

This table contains 9 rows and 2 columns.

wage: 9 non-missing numeric entries; minimum 0, maximum 526, mean 68.2.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 23. Outputs Part2 Individual T Tests

Source: /content/final_project_outputs/tables/outputs_part2_individual_t_tests.csv

Variable	Model	Coefficient	Standard error	t-statistic	p-value	95% CI lower	95% CI upper
const	Simple OLS	-0.9049	0.685	-1.321	0.1871	-2.2505	0.4408
educ	Simple OLS	0.5414	0.0532	10.1667	2.783e-22	0.4368	0.646
Intercept	Multiple OLS	-1.6022	0.7311	-2.1916	0.0289	-3.0384	-0.166
educ	Multiple OLS	0.5551	0.0501	11.0897	8.574e-26	0.4568	0.6534
exper	Multiple OLS	0.0187	0.012	1.557	0.1201	-0.0049	0.0424
tenure	Multiple OLS	0.1388	0.0212	6.5618	1.290e-10	0.0973	0.1804
female	Multiple OLS	-1.7424	0.2668	-6.5303	1.566e-10	-2.2666	-1.2182
married	Multiple OLS	0.5566	0.2867	1.941	0.0528	-0.0068	1.1199
nonwhite	Multiple OLS	-0.0658	0.4266	-0.1543	0.8775	-0.9038	0.7722

Display limited to first 100 rows and first 8 columns; complete data remain in the CSV/workbook.

Interpretation

This table contains 9 rows and 9 columns.

Estimated coefficient for const is -0.9049 (p-value 0.1871); not statistically significant at the 5% level under the reported model.

Estimated coefficient for educ is 0.5414 (p-value 2.783e-22); statistically significant at the 5% level under the reported model.

Estimated coefficient for Intercept is -1.6022 (p-value 0.02885); statistically significant at the 5% level under the reported model.

Estimated coefficient for educ is 0.5551 (p-value 8.574e-26); statistically significant at the 5% level under the reported model.

Estimated coefficient for exper is 0.0187 (p-value 0.1201); not statistically significant at the 5% level under the reported model.

Estimated coefficient for tenure is 0.1388 (p-value 1.29e-10); statistically significant at the 5% level under the reported model.

Estimated coefficient for female is -1.7424 (p-value 1.566e-10); statistically significant at the 5% level under the reported model.

Estimated coefficient for married is 0.5566 (p-value 0.0528); not statistically significant at the 5% level under the reported model.

Estimated coefficient for nonwhite is -0.0658 (p-value 0.8775); not statistically significant at the 5% level under the reported model.

Table 24. Outputs Part2 Joint F Test Experience Tenure

Source: /content/final_project_outputs/tables/outputs_part2_joint_f_test_experience_tenure.csv

Test	Null hypothesis	F-statistic	Numerator df	Denominator df	p-value	Reject H0 at 5%
Joint significance of experience and tenure	beta_exper = 0 and beta_tenure = 0	35.2061	2	519	4.594e-15	True

Interpretation

This table contains 1 rows and 7 columns.

Joint significance of experience and tenure: p-value = 4.594e-15; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Table 25. Outputs Part2 Model Sample Descriptive Statistics

Source: /content/final_project_outputs/tables/outputs_part2_model_sample_descriptive_statistics.csv

Unnamed: 0	count	mean	median	std	min	max
wage	526	5.8961	4.65	3.6931	0.53	24.98
educ	526	12.5627	12	2.769	0	18
exper	526	17.0171	13.5	13.5722	1	51
tenure	526	5.1046	2	7.2245	0	44
female	526	0.4791	0	0.5	0	1
married	526	0.6084	1	0.4886	0	1
nonwhite	526	0.1027	0	0.3038	0	1

Interpretation

This table contains 7 rows and 7 columns.

count: 7 non-missing numeric entries; minimum 526, maximum 526, mean 526.

mean: 7 non-missing numeric entries; minimum 0.1027, maximum 17.02, mean 5.967.

median: 7 non-missing numeric entries; minimum 0, maximum 13.5, mean 4.736.

std: 7 non-missing numeric entries; minimum 0.3038, maximum 13.57, mean 4.079.

min: 7 non-missing numeric entries; minimum 0, maximum 1, mean 0.2186.

max: 7 non-missing numeric entries; minimum 1, maximum 51, mean 20.14.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 26. Outputs Part2 Multiple Model Interpretation

Source: /content/final_project_outputs/tables/outputs_part2_multiple_model_interpretation.csv

Variable	Estimated coefficient	p-value	Statistical significance at 5%	Interpretation
educ	0.5551	8.574e-26	Statistically significant	Holding the other included regressors constant, a one-unit increase in educ is associated with an estimated 0.5551-unit change in hourly wage. This is an association, not necessarily a causal effect.
exper	0.0187	0.1201	Not statistically significant	Holding the other included regressors constant, a one-unit increase in exper is associated with an estimated 0.0187-unit change in hourly wage. This is an association, not necessarily a causal effect.
tenure	0.1388	1.290e-10	Statistically significant	Holding the other included regressors constant, a one-unit increase in tenure is associated with an estimated 0.1388-unit change in hourly wage. This is an association, not necessarily a causal effect.

Variable	Estimated coefficient	p-value	Statistical significance at 5%	Interpretation
female	-1.7424	1.566e-10	Statistically significant	Holding the other included regressors constant, a one-unit increase in female is associated with an estimated -1.7424-unit change in hourly wage. This is an association, not necessarily a causal effect.
married	0.5566	0.0528	Not statistically significant	Holding the other included regressors constant, a one-unit increase in married is associated with an estimated 0.5566-unit change in hourly wage. This is an association, not necessarily a causal effect.
nonwhite	-0.0658	0.8775	Not statistically significant	Holding the other included regressors constant, a one-unit increase in nonwhite is associated with an estimated -0.0658-unit change in hourly wage. This is an association, not necessarily a causal effect.

Interpretation

This table contains 6 rows and 5 columns.

educ: p-value = 8.574e-26; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

exper: p-value = 0.1201; do not reject the null hypothesis at the 5% level, conditional on the test's assumptions.

tenure: p-value = 1.29e-10; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

female: p-value = 1.566e-10; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

married: p-value = 0.0528; do not reject the null hypothesis at the 5% level, conditional on the test's assumptions.

nonwhite: p-value = 0.8775; do not reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Table 27. Outputs Part2 Ols Coefficient Comparison

Source: /content/final_project_outputs/tables/outputs_part2_ols_coefficient_comparison.csv

Model	Variable	Coefficient	Standard error	t-statistic	p-value	95% CI lower	95% CI upper
Simple OLS	const	-0.9049	0.685	-1.321	0.1871	-2.2505	0.4408
Simple OLS	educ	0.5414	0.0532	10.1667	2.783e-22	0.4368	0.646
Multiple OLS	Intercept	-1.6022	0.7311	-2.1916	0.0289	-3.0384	-0.166
Multiple OLS	educ	0.5551	0.0501	11.0897	8.574e-26	0.4568	0.6534
Multiple OLS	exper	0.0187	0.012	1.557	0.1201	-0.0049	0.0424
Multiple OLS	tenure	0.1388	0.0212	6.5618	1.290e-10	0.0973	0.1804
Multiple OLS	female	-1.7424	0.2668	-6.5303	1.566e-10	-2.2666	-1.2182
Multiple OLS	married	0.5566	0.2867	1.941	0.0528	-0.0068	1.1199
Multiple OLS	nonwhite	-0.0658	0.4266	-0.1543	0.8775	-0.9038	0.7722

Interpretation

This table contains 9 rows and 8 columns.

Estimated coefficient for const is -0.9049 (p-value 0.1871); not statistically significant at the 5% level under the reported model.

Estimated coefficient for educ is 0.5414 (p-value 2.783e-22); statistically significant at the 5% level under the reported model.

Estimated coefficient for Intercept is -1.6022 (p-value 0.02885); statistically significant at the 5% level under the reported model.

Estimated coefficient for educ is 0.5551 (p-value 8.574e-26); statistically significant at the 5% level under the reported model.

Estimated coefficient for exper is 0.0187 (p-value 0.1201); not statistically significant at the 5% level under the reported model.

Estimated coefficient for tenure is 0.1388 (p-value 1.29e-10); statistically significant at the 5% level under the reported model.

Estimated coefficient for female is -1.7424 (p-value 1.566e-10); statistically significant at the 5% level under the reported model.

Estimated coefficient for married is 0.5566 (p-value 0.0528); not statistically significant at the 5% level under the reported model.

Estimated coefficient for nonwhite is -0.0658 (p-value 0.8775); not statistically significant at the 5% level under the reported model.

Table 28. Outputs Part2 Ols Model Fit Comparison

Source: /content/final_project_outputs/tables/outputs_part2_ols_model_fit_comparison.csv

Model	Dependent variable	Observations	R-squared	Adjusted R-squared	Residual standard error	F-statistic	F-test p-value
Simple OLS	wage	526	0.1648	0.1632	3.3784	103.3627	2.783e-22
Multiple OLS	wage	526	0.3682	0.3609	2.9524	50.4142	8.269e-49

Interpretation

This table contains 2 rows and 8 columns.

Simple OLS: F-test p-value = 2.783e-22; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Multiple OLS: F-test p-value = 8.269e-49; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Table 29. Outputs Part2 Overall F Tests

Source: /content/final_project_outputs/tables/outputs_part2_overall_f_tests.csv

Model	Null hypothesis	F-statistic	Numerator df	Denominator df	p-value	Reject H0 at 5%
Simple OLS	All slope coefficients equal zero	103.3627	1	524	2.783e-22	True
Multiple OLS	All slope coefficients equal zero	50.4142	6	519	8.269e-49	True

Interpretation

This table contains 2 rows and 7 columns.

Simple OLS: p-value = 2.783e-22; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Multiple OLS: p-value = 8.269e-49; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Table 30. Outputs Part2 Sample Summary

Source: /content/final_project_outputs/tables/outputs_part2_sample_summary.csv

Unnamed: 0	Initial observations	Complete observations	Observations excluded for missing/non-finite values
wage1 model sample	526	526	0

Interpretation

This table contains 1 rows and 4 columns.

Initial observations: 1 non-missing numeric entries; minimum 526, maximum 526, mean 526.

Complete observations: 1 non-missing numeric entries; minimum 526, maximum 526, mean 526.

Observations excluded for missing/non-finite values: 1 non-missing numeric entries; minimum 0, maximum 0, mean 0.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 31. Outputs Part3 Education Indicator Interaction

Source: /content/final_project_outputs/tables/outputs_part3_education_indicator_interaction.csv

model	term	coefficient	std_error	t_statistic	p_value	ci_lower_95	ci_upper_95
Log wage with education interaction	Intercept	0.4553	0.1235	3.6877	2.502e-04	0.2127	0.6978
Log wage with education interaction	educ	0.0866	0.0088	9.8706	3.587e-21	0.0694	0.1038
Log wage with education interaction	exper	0.0031	0.0017	1.8678	0.0623	-1.630e-04	0.0065
Log wage with education interaction	tenure	0.0169	0.003	5.7143	1.861e-08	0.0111	0.0228
Log wage with education interaction	female	-0.199	0.1731	-1.1496	0.2508	-0.5391	0.1411
Log wage with education interaction	married	0.1256	0.0401	3.129	0.0019	0.0467	0.2044
Log wage with education interaction	nonwhite	-5.823e-04	0.0598	-0.0097	0.9922	-0.1181	0.1169
Log wage with education interaction	educ:female	-0.0069	0.0135	-0.5121	0.6088	-0.0334	0.0196

Interpretation

This table contains 8 rows and 8 columns.

Estimated coefficient for Intercept is 0.4553 (p-value 0.0002502); statistically significant at the 5% level under the reported model.

Estimated coefficient for educ is 0.0866 (p-value 3.587e-21); statistically significant at the 5% level under the reported model.

Estimated coefficient for exper is 0.0031 (p-value 0.06235); not statistically significant at the 5% level under the reported model.

Estimated coefficient for tenure is 0.0169 (p-value 1.861e-08); statistically significant at the 5% level under the reported model.

Estimated coefficient for female is -0.1990 (p-value 0.2508); not statistically significant at the 5% level under the reported model.

Estimated coefficient for married is 0.1256 (p-value 0.001853); statistically significant at the 5% level under the reported model.

Estimated coefficient for nonwhite is -0.0006 (p-value 0.9922); not statistically significant at the 5% level under the reported model.

Estimated coefficient for educ:female is -0.0069 (p-value 0.6088); not statistically significant at the 5% level under the reported model.

Table 32. Outputs Part3 Functional Form Coefficients

Source: /content/final_project_outputs/tables/outputs_part3_functional_form_coefficients.csv

model	term	coefficient	std_error	t_statistic	p_value	ci_lower_95	ci_upper_95
Linear wage	Intercept	-1.6022	0.7311	-2.1916	0.0289	-3.0384	-0.166
Linear wage	educ	0.5551	0.0501	11.0897	8.574e-26	0.4568	0.6534
Linear wage	exper	0.0187	0.012	1.557	0.1201	-0.0049	0.0424
Linear wage	tenure	0.1388	0.0212	6.5618	1.290e-10	0.0973	0.1804
Linear wage	female	-1.7424	0.2668	-6.5303	1.566e-10	-2.2666	-1.2182
Linear wage	married	0.5566	0.2867	1.941	0.0528	-0.0068	1.1199
Linear wage	nonwhite	-0.0658	0.4266	-0.1543	0.8775	-0.9038	0.7722
Log wage	Intercept	0.4907	0.1022	4.7996	2.082e-06	0.2898	0.6915
Log wage	educ	0.0839	0.007	11.9839	2.198e-29	0.0701	0.0976
Log wage	exper	0.0031	0.0017	1.8614	0.0632	-1.736e-04	0.0064

model	term	coefficient	std_error	t_statistic	p_value	ci_lower_95	ci_upper_95
Log wage	tenure	0.0169	0.003	5.7017	1.993e-08	0.0111	0.0227
Log wage	female	-0.2856	0.0373	-7.6537	9.590e-14	-0.3589	-0.2123
Log wage	married	0.1256	0.0401	3.1333	0.0018	0.0469	0.2044
Log wage	nonwhite	-0.0025	0.0597	-0.0424	0.9662	-0.1197	0.1147

Interpretation

This table contains 14 rows and 8 columns.

Estimated coefficient for Intercept is -1.6022 (p-value 0.02885); statistically significant at the 5% level under the reported model.

Estimated coefficient for educ is 0.5551 (p-value 8.574e-26); statistically significant at the 5% level under the reported model.

Estimated coefficient for exper is 0.0187 (p-value 0.1201); not statistically significant at the 5% level under the reported model.

Estimated coefficient for tenure is 0.1388 (p-value 1.29e-10); statistically significant at the 5% level under the reported model.

Estimated coefficient for female is -1.7424 (p-value 1.566e-10); statistically significant at the 5% level under the reported model.

Estimated coefficient for married is 0.5566 (p-value 0.0528); not statistically significant at the 5% level under the reported model.

Estimated coefficient for nonwhite is -0.0658 (p-value 0.8775); not statistically significant at the 5% level under the reported model.

Estimated coefficient for Intercept is 0.4907 (p-value 2.082e-06); statistically significant at the 5% level under the reported model.

Estimated coefficient for educ is 0.0839 (p-value 2.198e-29); statistically significant at the 5% level under the reported model.

Estimated coefficient for exper is 0.0031 (p-value 0.06325); not statistically significant at the 5% level under the reported model.

Estimated coefficient for tenure is 0.0169 (p-value 1.993e-08); statistically significant at the 5% level under the reported model.

Estimated coefficient for female is -0.2856 (p-value 9.59e-14); statistically significant at the 5% level under the reported model.

Estimated coefficient for married is 0.1256 (p-value 0.001826); statistically significant at the 5% level under the reported model.

Estimated coefficient for nonwhite is -0.0025 (p-value 0.9662); not statistically significant at the 5% level under the reported model.

Table 33. Outputs Part3 Functional Form Model Summary

Source: /content/final_project_outputs/tables/outputs_part3_functional_form_model_summary.csv

model	dependent_variable	n	r_squared	adjusted_r_squared	aic
Linear wage	wage	526	0.3682	0.3609	2638.5762
Log wage	log_wage	526	0.4036	0.3967	569.0077

Interpretation

This table contains 2 rows and 6 columns.

n: 2 non-missing numeric entries; minimum 526, maximum 526, mean 526.

r_squared: 2 non-missing numeric entries; minimum 0.3682, maximum 0.4036, mean 0.3859.

adjusted_r_squared: 2 non-missing numeric entries; minimum 0.3609, maximum 0.3967, mean 0.3788.

aic: 2 non-missing numeric entries; minimum 569, maximum 2639, mean 1604.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 34. Outputs Part3 Joint Dummy F Test

Source: /content/final_project_outputs/tables/outputs_part3_joint_dummy_f_test.csv

test	null	f_statistic	df_num	df_denom	p_value	reject_at_5pct
Joint F-test: worker-characteristic indicators	female = married = nonwhite = 0	25.411	3	519	2.366e-15	True

Interpretation

This table contains 1 rows and 7 columns.

Joint F-test: worker-characteristic indicators: p_value = 2.366e-15; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Table 35. Outputs Part3 Log Wage Coefficient Interpretation

Source: /content/final_project_outputs/tables/outputs_part3_log_wage_coefficient_interpretation.csv

variable	coefficient	implied_percent_difference	indicator
educ	0.0839	8.7501	False
exper	0.0031	0.3139	False
tenure	0.0169	1.7012	False
female	-0.2856	-24.8413	True
married	0.1256	13.387	True
nonwhite	-0.0025	-0.2524	True

Interpretation

This table contains 6 rows and 4 columns.

Estimated coefficient for educ is 0.0839.

Estimated coefficient for exper is 0.0031.

Estimated coefficient for tenure is 0.0169.

Estimated coefficient for female is -0.2856.

Estimated coefficient for married is 0.1256.

Estimated coefficient for nonwhite is -0.0025.

Table 36. Outputs Part3 Mwd Functional Form Test

Source: /content/final_project_outputs/tables/outputs_part3_mwd_functional_form_test.csv

status	reason
Not computed	Nonpositive or nonfinite fitted wages prevent log(fitted wage).

Interpretation

This table contains 1 rows and 2 columns.

This is a categorical/text summary table; interpret its entries using the row and column labels.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 37. Outputs Part4 Conventional Vs Robust Standard Errors

Source: /content/final_project_outputs/tables/outputs_part4_conventional_vs_robust_standard_errors.csv

Variable	Coefficient	Conventional standard error	Robust standard error (HC1)	Conventional p-value	Robust p-value (HC1)
Intercept	-1.6022	0.7311	0.8227	0.0289	0.052
educ	0.5551	0.0501	0.0626	8.574e-26	1.158e-17
exper	0.0187	0.012	0.0101	0.1201	0.065
tenure	0.1388	0.0212	0.028	1.290e-10	9.613e-07
female	-1.7424	0.2668	0.2504	1.566e-10	1.038e-11
married	0.5566	0.2867	0.2593	0.0528	0.0323
nonwhite	-0.0658	0.4266	0.3888	0.8775	0.8657

Interpretation

This table contains 7 rows and 6 columns.

Estimated coefficient for Intercept is -1.6022.

Estimated coefficient for educ is 0.5551.

Estimated coefficient for exper is 0.0187.

Estimated coefficient for tenure is 0.1388.

Estimated coefficient for female is -1.7424.

Estimated coefficient for married is 0.5566.

Estimated coefficient for nonwhite is -0.0658.

Table 38. Outputs Part4 Diagnostic Summary

Source: /content/final_project_outputs/tables/outputs_part4_diagnostic_summary.csv

Model	Observations	R-squared	Adjusted R-squared	Breusch-Pagan-Godfrey p-value	White test p-value	Jarque-Bera p-value
Multiple OLS: wage ~ educ + exper + tenure + female + married + nonwhite	526	0.3682	0.3609	6.701e-08	5.624e-07	8.109e-159

Interpretation

This table contains 1 rows and 7 columns.

Multiple OLS: wage ~ educ + exper + tenure + female + married + nonwhite: Breusch-Pagan-Godfrey p-value = 6.701e-08; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Multiple OLS: wage ~ educ + exper + tenure + female + married + nonwhite: White test p-value = 5.624e-07; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Multiple OLS: wage ~ educ + exper + tenure + female + married + nonwhite: Jarque-Bera p-value = 8.109e-159; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Table 39. Outputs Part4 Heteroskedasticity Tests

Source: /content/final_project_outputs/tables/outputs_part4_heteroskedasticity_tests.csv

Test	LM statistic	LM p-value	F statistic	F p-value	Reject homoskedasticity at 5%
Breusch-Pagan-Godfrey	44.2148	6.701e-08	7.9384	3.329e-08	True
White	73.8515	5.624e-07	3.4096	1.552e-07	True

Interpretation

This table contains 2 rows and 6 columns.

Breusch-Pagan-Godfrey: LM p-value = 6.701e-08; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Breusch-Pagan-Godfrey: F p-value = 3.329e-08; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

White: LM p-value = 5.624e-07; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

White: F p-value = 1.552e-07; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Table 40. Outputs Part4 Jarque Bera Normality Test

Source: /content/final_project_outputs/tables/outputs_part4_jarque_bera_normality_test.csv

Test	Statistic	p-value	Reject normality at 5%	Number of residuals
Jarque-Bera	728.0361	8.109e-159	True	526

Interpretation

This table contains 1 rows and 5 columns.

Jarque-Bera: p-value = 8.109e-159; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Table 41. Outputs Part4 Predictor Correlation Matrix

Source: /content/final_project_outputs/tables/outputs_part4_predictor_correlation_matrix.csv

Unnamed: 0	educ	exper	tenure	female	married	nonwhite
educ	1	-0.2995	-0.0562	-0.085	0.0689	-0.0847
exper	-0.2995	1	0.4993	-0.0416	0.317	0.0144
tenure	-0.0562	0.4993	1	-0.1979	0.2399	0.0116
female	-0.085	-0.0416	-0.1979	1	-0.1661	-0.0109
married	0.0689	0.317	0.2399	-0.1661	1	-0.0623
nonwhite	-0.0847	0.0144	0.0116	-0.0109	-0.0623	1

Interpretation

This table contains 6 rows and 7 columns.

educ: 6 non-missing numeric entries; minimum -0.2995, maximum 1, mean 0.09058.

exper: 6 non-missing numeric entries; minimum -0.2995, maximum 1, mean 0.2482.

tenure: 6 non-missing numeric entries; minimum -0.1979, maximum 1, mean 0.2494.

female: 6 non-missing numeric entries; minimum -0.1979, maximum 1, mean 0.08306.

married: 6 non-missing numeric entries; minimum -0.1661, maximum 1, mean 0.2329.

nonwhite: 6 non-missing numeric entries; minimum -0.08465, maximum 1, mean 0.1447.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 42. Outputs Part4 Variance Inflation Factors

Source: /content/final_project_outputs/tables/outputs_part4_variance_inflation_factors.csv

Variable	VIF
educ	1.1571
exper	1.6084
tenure	1.4072
female	1.0722
married	1.1822
nonwhite	1.0115

Interpretation

This table contains 6 rows and 2 columns.

VIF: 6 non-missing numeric entries; minimum 1.012, maximum 1.608, mean 1.24.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 43. Outputs Part5 Autocorrelation Tests

Source: /content/final_project_outputs/tables/outputs_part5_autocorrelation_tests.csv

Test	Statistic	p-value	Lag order	Interpretation
Durbin-Watson	0.8015		1	The Durbin-Watson statistic is descriptive; use its appropriate critical-value bounds or a suitable formal procedure for inference.
Breusch-Godfrey	26.0731	3.059e-05	4	Tests the null of no residual autocorrelation up to the specified lag order.

Interpretation

This table contains 2 rows and 5 columns.

Durbin-Watson: p-value = nan; do not reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Breusch-Godfrey: p-value = 3.059e-05; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Table 44. Outputs Part5 Breusch Godfrey F Test

Source: /content/final_project_outputs/tables/outputs_part5_breusch_godfrey_f_test.csv

Test	F-statistic	p-value	Lag order
Breusch-Godfrey F form	10.8903	1.989e-06	4

Interpretation

This table contains 1 rows and 4 columns.

Breusch-Godfrey F form: p-value = 1.989e-06; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Table 45. Outputs Part5 Joint F Test Selected Regressors

Source: /content/final_project_outputs/tables/outputs_part5_joint_f_test_selected_regressors.csv

Test	Restrictions	F-statistic	Numerator df	Denominator df	p-value	Reject null at 5%
Joint F-test for selected regressors	married = 0 and nonwhite = 0	1.9208	2	519	0.1475	False

Interpretation

This table contains 1 rows and 7 columns.

Joint F-test for selected regressors: p-value = 0.1475; do not reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Table 46. Outputs Part5 Phillips Time Series Data Notes

Source: /content/final_project_outputs/tables/outputs_part5_phillips_time_series_data_notes.csv

Observations used	Dependent variable	Explanatory variable	Ordering	Time index verified from metadata	Interpretation note
56	inf	unem	Original dataset row order	False	Autocorrelation tests require meaningful temporal ordering. Confirm the dataset's documented time index before interpreting these results as time-series evidence.

Interpretation

This table contains 1 rows and 6 columns.

Observations used: 1 non-missing numeric entries; minimum 56, maximum 56, mean 56.

Dependent variable: 1 non-missing numeric entries; minimum inf, maximum inf, mean inf.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 47. Outputs Part5 Ramsey Reset Test

Source: /content/final_project_outputs/tables/outputs_part5_ramsey_reset_test.csv

Test	Specification	F-statistic	p-value	Reject null at 5%
Ramsey RESET	Powers of fitted values through cubic	15.1683	3.968e-07	True

Interpretation

This table contains 1 rows and 5 columns.

Ramsey RESET: p-value = 3.968e-07; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Table 48. Data Preparation Decisions

Source: /content/outputs_part1/data_preparation_decisions.csv

Item	Decision
Row deletion (both datasets)	No observations are removed. Unusual values are flagged, not treated as errors.
Outliers in wage	Retained. High wages are plausible in survey data; any robustness treatment is deferred and would be justified explicitly.
Missing values - wage1	0 missing cells found. No imputation needed.
Missing values - phillips	4 missing cells found, all in lagged/differenced variables for the first year (by construction). Not imputed; Part 5 will use the lags it actually needs.
Duplicate rows	wage1 has 0 and phillips has 0 exact duplicate rows. Not deleted (no ID variable to establish true duplication).
New variable: log_wage	Created as natural log of wage for strictly positive wages. Observations with wage <= 0 would receive NaN (currently 0). Original 'wage' and 'lwage' are preserved.
Existing transformed variables	lwage, expersq and tenursq (supplied by the dataset) are kept unchanged; they are verified against the originals in Cell 9.

Interpretation

This table contains 7 rows and 2 columns.

This is a categorical/text summary table; interpret its entries using the row and column labels.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 49. Dataset Summary

Source: /content/outputs_part1/dataset_summary.csv

Dataset	Observations	Variables	Numeric variables	Missing cells	Duplicate rows	Infinite values	Time information
wage1	526	24	24	0	0	0	none (cross-section)
phillips	56	7	7	4	0	0	annual, 1948-2003

Display limited to first 100 rows and first 8 columns; complete data remain in the CSV/workbook.

Interpretation

This table contains 2 rows and 9 columns.

wage1: 526 observations and 24 variables.

phillips: 56 observations and 7 variables.

These counts describe the datasets used for the project's cross-sectional wage analysis and supplementary time-series analysis.

Table 50. Phillips Audit Summary

Source: /content/outputs_part1/phillips_audit_summary.csv

metric	value
n_obs	56
n_vars	7

metric	value
missing_cells	4
missing_by_col	{'inf_1': 1, 'unem_1': 1, 'cinf': 1, 'cunem': 1}
n_duplicate_rows	0
n_inf	0
non_numeric_cols	[]
numeric_cols	['year', 'unem', 'inf', 'inf_1', 'unem_1', 'cinf', 'cunem']
flag_counts	{'unem < 0 or unem > 100': 0, '|inf| > 20 (heuristic screen)': 0}

Interpretation

This table contains 9 rows and 2 columns.

This is a categorical/text summary table; interpret its entries using the row and column labels.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 51. Phillips Derived Variable Checks

Source: /content/outputs_part1/phillips_derived_variable_checks.csv

Unnamed: 0	max absolute difference
inf_1 vs inf shifted by one year	0
unem_1 vs unem shifted by one year	0
cinf vs inf - inf_1	2.384e-07
cunem vs unem - unem_1	1.776e-15

Interpretation

This table contains 4 rows and 2 columns.

max absolute difference: 4 non-missing numeric entries; minimum 0, maximum 2.384e-07, mean 5.96e-08.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 52. Phillips Descriptive Statistics

Source: /content/outputs_part1/phillips_descriptive_statistics.csv

Unnamed: 0	documented meaning	count	mean	median	std	min	max
unem	civilian unemployment rate, %	56	5.6339	5.55	1.5088	2.9	9.7
inf	percentage change in CPI	56	3.8839	3.05	3.0404	-1.2	13.5
cunem	unem - unem_1	55	0.04	-0.2	1.0642	-2.1	2.9
cinf	inf - inf_1	55	-0.1055	0.2	2.414	-9.3	6.6

Interpretation

This table contains 4 rows and 8 columns.

count: 4 non-missing numeric entries; minimum 55, maximum 56, mean 55.5.

mean: 4 non-missing numeric entries; minimum -0.1055, maximum 5.634, mean 2.363.

median: 4 non-missing numeric entries; minimum -0.2, maximum 5.55, mean 2.15.

std: 4 non-missing numeric entries; minimum 1.064, maximum 3.04, mean 2.007.

min: 4 non-missing numeric entries; minimum -9.3, maximum 2.9, mean -2.425.

max: 4 non-missing numeric entries; minimum 2.9, maximum 13.5, mean 8.175.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 53. Phillips Numeric Integrity

Source: /content/outputs_part1/phillips_numeric_integrity.csv

variable	numeric_dtype	entries_not_convertible_to_number	infinite_values	missing_values
year	True	0	0	0
unem	True	0	0	0
inf	True	0	0	0
inf_1	True	0	0	1
unem_1	True	0	0	1
cinf	True	0	0	1
cunem	True	0	0	1

Interpretation

This table contains 7 rows and 5 columns.

entries_not_convertible_to_number: 7 non-missing numeric entries; minimum 0, maximum 0, mean 0.

infinite_values: 7 non-missing numeric entries; minimum 0, maximum 0, mean 0.

missing_values: 7 non-missing numeric entries; minimum 0, maximum 1, mean 0.5714.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 54. Phillips Time Series Checks

Source: /content/outputs_part1/phillips_time_series_checks.csv

Unnamed: 0	check	passed
0	year_whole_numbers	True
1	year_sorted	True
2	year_unique	True
3	year_consecutive	True
4	derived_variables_consistent	True

Interpretation

This table contains 5 rows and 3 columns.

Unnamed: 0: 5 non-missing numeric entries; minimum 0, maximum 4, mean 2.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 55. Phillips Working Data

Source: /content/outputs_part1/phillips_working_data.csv

year	unem	inf	inf_1	unem_1	cinf	cunem
1948	3.8	8.1				
1949	5.9	-1.2	8.1	3.8	-9.3	2.1
1950	5.3	1.3	-1.2	5.9	2.5	-0.6
1951	3.3	7.9	1.3	5.3	6.6	-2
1952	3	1.9	7.9	3.3	-6	-0.3
1953	2.9	0.8	1.9	3	-1.1	-0.1
1954	5.5	0.7	0.8	2.9	-0.1	2.6

year	unem	inf	inf_1	unem_1	cinf	cunem
1955	4.4	-0.4	0.7	5.5	-1.1	-1.1
1956	4.1	1.5	-0.4	4.4	1.9	-0.3
1957	4.3	3.3	1.5	4.1	1.8	0.2
1958	6.8	2.8	3.3	4.3	-0.5	2.5
1959	5.5	0.7	2.8	6.8	-2.1	-1.3
1960	5.5	1.7	0.7	5.5	1	0
1961	6.7	1	1.7	5.5	-0.7	1.2
1962	5.5	1	1	6.7	0	-1.2
1963	5.7	1.3	1	5.5	0.3	0.2
1964	5.2	1.3	1.3	5.7	0	-0.5
1965	4.5	1.6	1.3	5.2	0.3	-0.7
1966	3.8	2.9	1.6	4.5	1.3	-0.7
1967	3.8	3.1	2.9	3.8	0.2	0
1968	3.6	4.2	3.1	3.8	1.1	-0.2
1969	3.5	5.5	4.2	3.6	1.3	-0.1
1970	4.9	5.7	5.5	3.5	0.2	1.4
1971	5.9	4.4	5.7	4.9	-1.3	1
1972	5.6	3.2	4.4	5.9	-1.2	-0.3
1973	4.9	6.2	3.2	5.6	3	-0.7
1974	5.6	11	6.2	4.9	4.8	0.7
1975	8.5	9.1	11	5.6	-1.9	2.9
1976	7.7	5.8	9.1	8.5	-3.3	-0.8
1977	7.1	6.5	5.8	7.7	0.7	-0.6
1978	6.1	7.6	6.5	7.1	1.1	-1
1979	5.8	11.3	7.6	6.1	3.7	-0.3
1980	7.1	13.5	11.3	5.8	2.2	1.3
1981	7.6	10.3	13.5	7.1	-3.2	0.5
1982	9.7	6.2	10.3	7.6	-4.1	2.1
1983	9.6	3.2	6.2	9.7	-3	-0.1
1984	7.5	4.3	3.2	9.6	1.1	-2.1
1985	7.2	3.6	4.3	7.5	-0.7	-0.3
1986	7	1.9	3.6	7.2	-1.7	-0.2
1987	6.2	3.6	1.9	7	1.7	-0.8
1988	5.5	4.1	3.6	6.2	0.5	-0.7
1989	5.3	4.8	4.1	5.5	0.7	-0.2
1990	5.6	5.4	4.8	5.3	0.6	0.3
1991	6.8	4.2	5.4	5.6	-1.2	1.2
1992	7.5	3	4.2	6.8	-1.2	0.7
1993	6.9	3	3	7.5	0	-0.6
1994	6.1	2.6	3	6.9	-0.4	-0.8
1995	5.6	2.8	2.6	6.1	0.2	-0.5
1996	5.4	3	2.8	5.6	0.2	-0.2
1997	4.9	2.3	3	5.4	-0.7	-0.5
1998	4.5	1.6	2.3	4.9	-0.7	-0.4

year	unem	inf	inf_1	unem_1	cinf	cunem
1999	4.2	2.2	1.6	4.5	0.6	-0.3
2000	4	3.4	2.2	4.2	1.2	-0.2
2001	4.8	2.8	3.4	4	-0.6	0.8
2002	5.8	1.6	2.8	4.8	-1.2	1
2003	6	2.3	1.6	5.8	0.7	0.2

Interpretation

This table contains 56 rows and 7 columns.

year: 56 non-missing numeric entries; minimum 1948, maximum 2003, mean 1976.

unem: 56 non-missing numeric entries; minimum 2.9, maximum 9.7, mean 5.634.

inf: 56 non-missing numeric entries; minimum -1.2, maximum 13.5, mean 3.884.

inf_1: 55 non-missing numeric entries; minimum -1.2, maximum 13.5, mean 3.913.

unem_1: 55 non-missing numeric entries; minimum 2.9, maximum 9.7, mean 5.627.

cinf: 55 non-missing numeric entries; minimum -9.3, maximum 6.6, mean -0.1055.

cunem: 55 non-missing numeric entries; minimum -2.1, maximum 2.9, mean 0.04.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 56. Wage1 Audit Summary

Source: /content/outputs_part1/wage1_audit_summary.csv

metric	value
n_obs	526
n_vars	24
missing_cells	0
missing_by_col	{}
n_duplicate_rows	0
n_inf	0
non_numeric_cols	[]
numeric_cols	['wage', 'educ', 'exper', 'tenure', 'nonwhite', 'female', 'married', 'numdep', 'smsa', 'northcen', 'south', 'west', 'construc', 'ndurman', 'trcommpu', 'trade', 'services', 'profserv', 'profocc', 'clerocc', 'servocc', 'lwage', 'expersq', 'tenursq']
flag_counts	{'wage <= 0': 0, 'educ < 0 or educ > 20': 0, 'exper < 0 or exper > 60': 0, 'tenure < 0 or tenure > 50': 0, 'tenure > exper (tenure exceeds potential experience)': 16, 'indicator value outside {0,1}': 0, 'lwage differs from log(wage) (tolerance 1e-6)': 0, 'expersq differs from exper^2': 0, 'tenursq differs from tenure^2': 0}

Interpretation

This table contains 9 rows and 2 columns.

This is a categorical/text summary table; interpret its entries using the row and column labels.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 57. Wage1 Correlation Matrix

Source: /content/outputs_part1/wage1_correlation_matrix.csv

Unnamed: 0	wage	log_wage	educ	exper	tenure	female	married
wage	1	0.9371	0.4059	0.1129	0.3469	-0.3401	0.2288

Unnamed: 0	wage	log_wage	educ	exper	tenure	female	married
log_wage	0.9371	1	0.4311	0.1114	0.3255	-0.3737	0.2707
educ	0.4059	0.4311	1	-0.2995	-0.0562	-0.085	0.0689
exper	0.1129	0.1114	-0.2995	1	0.4993	-0.0416	0.317
tenure	0.3469	0.3255	-0.0562	0.4993	1	-0.1979	0.2399
female	-0.3401	-0.3737	-0.085	-0.0416	-0.1979	1	-0.1661
married	0.2288	0.2707	0.0689	0.317	0.2399	-0.1661	1
nonwhite	-0.0385	-0.0389	-0.0847	0.0144	0.0116	-0.0109	-0.0623

Display limited to first 100 rows and first 8 columns; complete data remain in the CSV/workbook.

Interpretation

This table contains 8 rows and 9 columns.

wage: 8 non-missing numeric entries; minimum -0.3401, maximum 1, mean 0.3316.

log_wage: 8 non-missing numeric entries; minimum -0.3737, maximum 1, mean 0.3329.

educ: 8 non-missing numeric entries; minimum -0.2995, maximum 1, mean 0.1726.

exper: 8 non-missing numeric entries; minimum -0.2995, maximum 1, mean 0.2142.

tenure: 8 non-missing numeric entries; minimum -0.1979, maximum 1, mean 0.2711.

female: 8 non-missing numeric entries; minimum -0.3737, maximum 1, mean -0.02692.

married: 8 non-missing numeric entries; minimum -0.1661, maximum 1, mean 0.2371.

nonwhite: 8 non-missing numeric entries; minimum -0.08465, maximum 1, mean 0.09884.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 58. Wage1 Descriptive Statistics

Source: /content/outputs_part1/wage1_descriptive_statistics.csv

Unnamed: 0	documented meaning	count	mean	median	std	min	max
wage	average hourly earnings	526	5.8961	4.65	3.6931	0.53	24.98
educ	years of education	526	12.5627	12	2.769	0	18
exper	years potential experience	526	17.0171	13.5	13.5722	1	51
tenure	years with current employer	526	5.1046	2	7.2245	0	44
female	=1 if female	526	0.4791	0	0.5	0	1
married	=1 if married	526	0.6084	1	0.4886	0	1
nonwhite	=1 if nonwhite	526	0.1027	0	0.3038	0	1

Interpretation

This table contains 7 rows and 8 columns.

count: 7 non-missing numeric entries; minimum 526, maximum 526, mean 526.

mean: 7 non-missing numeric entries; minimum 0.1027, maximum 17.02, mean 5.967.

median: 7 non-missing numeric entries; minimum 0, maximum 13.5, mean 4.736.

std: 7 non-missing numeric entries; minimum 0.3038, maximum 13.57, mean 4.079.

min: 7 non-missing numeric entries; minimum 0, maximum 1, mean 0.2186.

max: 7 non-missing numeric entries; minimum 1, maximum 51, mean 20.14.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 59. Wage1 Group Characteristics

Source: /content/outputs_part1/wage1_group_characteristics.csv

female	married	educ	exper	tenure
0	0	12.5465	10.7093	3.2558
0	1	12.8989	20.6915	7.9468
1	0	12.1667	12.3417	2.725
1	1	12.4545	20.1439	4.4242

Interpretation

This table contains 4 rows and 5 columns.

female: 4 non-missing numeric entries; minimum 0, maximum 1, mean 0.5.

married: 4 non-missing numeric entries; minimum 0, maximum 1, mean 0.5.

educ: 4 non-missing numeric entries; minimum 12.17, maximum 12.9, mean 12.52.

exper: 4 non-missing numeric entries; minimum 10.71, maximum 20.69, mean 15.97.

tenure: 4 non-missing numeric entries; minimum 2.725, maximum 7.947, mean 4.588.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 60. Wage1 Group Gender

Source: /content/outputs_part1/wage1_group_gender.csv

female	count	mean	median	std	min	max
0	274	7.0995	6	4.1609	1.5	24.98
1	252	4.5877	3.75	2.5294	0.53	21.63

Interpretation

This table contains 2 rows and 7 columns.

female: 2 non-missing numeric entries; minimum 0, maximum 1, mean 0.5.

count: 2 non-missing numeric entries; minimum 252, maximum 274, mean 263.

mean: 2 non-missing numeric entries; minimum 4.588, maximum 7.099, mean 5.844.

median: 2 non-missing numeric entries; minimum 3.75, maximum 6, mean 4.875.

std: 2 non-missing numeric entries; minimum 2.529, maximum 4.161, mean 3.345.

min: 2 non-missing numeric entries; minimum 0.53, maximum 1.5, mean 1.015.

max: 2 non-missing numeric entries; minimum 21.63, maximum 24.98, mean 23.3.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 61. Wage1 Group Gender Marital

Source: /content/outputs_part1/wage1_group_gender_marital.csv

female	married	count	mean	median	std	min	max
0	0	86	5.168	4.41	2.7228	1.5	12.5
0	1	188	7.983	6.5	4.405	2.23	24.98
1	0	120	4.6116	3.5	2.9956	0.53	21.63
1	1	132	4.5659	4	2.0263	1.43	15

Interpretation

This table contains 4 rows and 8 columns.

female: 4 non-missing numeric entries; minimum 0, maximum 1, mean 0.5.

married: 4 non-missing numeric entries; minimum 0, maximum 1, mean 0.5.

count: 4 non-missing numeric entries; minimum 86, maximum 188, mean 131.5.

mean: 4 non-missing numeric entries; minimum 4.566, maximum 7.983, mean 5.582.

median: 4 non-missing numeric entries; minimum 3.5, maximum 6.5, mean 4.603.

std: 4 non-missing numeric entries; minimum 2.026, maximum 4.405, mean 3.037.

min: 4 non-missing numeric entries; minimum 0.53, maximum 2.23, mean 1.422.

max: 4 non-missing numeric entries; minimum 12.5, maximum 24.98, mean 18.53.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 62. Wage1 Group Marital Status

Source: /content/outputs_part1/wage1_group_marital_status.csv

married	count	mean	median	std	min	max
0	206	4.8439	3.75	2.8911	0.53	21.63
1	320	6.5735	5.49	3.9874	1.43	24.98

Interpretation

This table contains 2 rows and 7 columns.

married: 2 non-missing numeric entries; minimum 0, maximum 1, mean 0.5.

count: 2 non-missing numeric entries; minimum 206, maximum 320, mean 263.

mean: 2 non-missing numeric entries; minimum 4.844, maximum 6.573, mean 5.709.

median: 2 non-missing numeric entries; minimum 3.75, maximum 5.49, mean 4.62.

std: 2 non-missing numeric entries; minimum 2.891, maximum 3.987, mean 3.439.

min: 2 non-missing numeric entries; minimum 0.53, maximum 1.43, mean 0.98.

max: 2 non-missing numeric entries; minimum 21.63, maximum 24.98, mean 23.3.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 63. Wage1 Indicator Frequencies

Source: /content/outputs_part1/wage1_indicator_frequencies.csv

Unnamed: 0	variable	documented	count_0	count_1	share_1_%
0	nonwhite	=1 if nonwhite	472	54	10.2662
1	female	=1 if female	274	252	47.9087
2	married	=1 if married	206	320	60.8365
3	smsa	=1 if live in SMSA	146	380	72.2433
4	northcen	=1 if live in north central U.S	394	132	25.0951
5	south	=1 if live in southern region	339	187	35.5513
6	west	=1 if live in western region	437	89	16.9202
7	construc	=1 if work in construc. indus.	502	24	4.5627
8	ndurman	=1 if in nondur. manuf. indus.	466	60	11.4068
9	trcommpu	=1 if in trans, commun, pub ut	503	23	4.3726

Unnamed: 0	variable	documented	count_0	count_1	share_1_%
10	trade	=1 if in wholesale or retail	375	151	28.7072
11	services	=1 if in services indus.	473	53	10.076
12	profserv	=1 if in prof. serv. indus.	390	136	25.8555
13	profocc	=1 if in profess. occupation	333	193	36.692
14	clerocc	=1 if in clerical occupation	438	88	16.73
15	servocc	=1 if in service occupation	452	74	14.0684

Interpretation

This table contains 16 rows and 6 columns.

Unnamed: 0: 16 non-missing numeric entries; minimum 0, maximum 15, mean 7.5.

count_0: 16 non-missing numeric entries; minimum 146, maximum 503, mean 387.5.

count_1: 16 non-missing numeric entries; minimum 23, maximum 380, mean 138.5.

share_1_%: 16 non-missing numeric entries; minimum 4.373, maximum 72.24, mean 26.33.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 64. Wage1 Numeric Integrity

Source: /content/outputs_part1/wage1_numeric_integrity.csv

variable	numeric_dtype	entries_not_convertible_to_number	infinite_values	missing_values
wage	True	0	0	0
educ	True	0	0	0
exper	True	0	0	0
tenure	True	0	0	0
nonwhite	True	0	0	0
female	True	0	0	0
married	True	0	0	0
numdep	True	0	0	0
smsa	True	0	0	0
northcen	True	0	0	0
south	True	0	0	0
west	True	0	0	0
construc	True	0	0	0
ndurman	True	0	0	0
trcommpu	True	0	0	0
trade	True	0	0	0
services	True	0	0	0
profserv	True	0	0	0
profocc	True	0	0	0
clerocc	True	0	0	0
servocc	True	0	0	0
lwage	True	0	0	0

variable	numeric_dtype	entries_not_convertible_to_number	infinite_values	missing_values
expersq	True	0	0	0
tenursq	True	0	0	0

Interpretation

This table contains 24 rows and 5 columns.

entries_not_convertible_to_number: 24 non-missing numeric entries; minimum 0, maximum 0, mean 0.

infinite_values: 24 non-missing numeric entries; minimum 0, maximum 0, mean 0.

missing_values: 24 non-missing numeric entries; minimum 0, maximum 0, mean 0.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 65. Wage1 Plausibility Flags

Source: /content/outputs_part1/wage1_plausibility_flags.csv

Unnamed: 0	rows_flagged
wage <= 0	0
educ < 0 or educ > 20	0
exper < 0 or exper > 60	0
tenure < 0 or tenure > 50	0
tenure > exper (tenure exceeds potential experience)	16
indicator value outside {0,1}	0
lwage differs from log(wage) (tolerance 1e-6)	0
expersq differs from exper^2	0
tenursq differs from tenure^2	0

Interpretation

This table contains 9 rows and 2 columns.

rows_flagged: 9 non-missing numeric entries; minimum 0, maximum 16, mean 1.778.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 66. Wage1 Variable Presence

Source: /content/outputs_part1/wage1_variable_presence.csv

Unnamed: 0	present	documented_meaning
wage	True	average hourly earnings
educ	True	years of education
exper	True	years potential experience
tenure	True	years with current employer
female	True	=1 if female
married	True	=1 if married
nonwhite	True	=1 if nonwhite

Interpretation

This table contains 7 rows and 3 columns.

This is a categorical/text summary table; interpret its entries using the row and column labels.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 67. Wage1 Working Data

Source: /content/outputs_part1/wage1_working_data.csv

wage	educ	exper	tenure	nonwhite	female	married	numdep
3.1	11	2	0	0	1	0	2
3.24	12	22	2	0	1	1	3
3	11	2	0	0	0	0	2
6	8	44	28	0	0	1	0
5.3	12	7	2	0	0	1	1
8.75	16	9	8	0	0	1	0
11.25	18	15	7	0	0	0	0
5	12	5	3	0	1	0	0
3.6	12	26	4	0	1	0	2
18.18	17	22	21	0	0	1	0
6.25	16	8	2	0	1	0	0
8.13	13	3	0	0	1	0	0
8.77	12	15	0	0	0	1	2
5.5	12	18	3	0	0	0	0
22.2	12	31	15	0	0	1	1
17.33	16	14	0	0	0	1	1
7.5	12	10	0	0	1	1	0
10.63	13	16	10	0	1	0	0
3.6	12	13	0	0	1	1	3
4.5	12	36	6	0	1	1	0
6.88	12	11	4	0	1	0	0
8.48	12	29	13	0	0	1	3
6.33	16	9	9	0	1	0	0
0.53	12	3	1	0	1	0	0
6	11	37	8	1	1	0	0
9.56	16	3	3	1	0	1	1
7.78	16	11	10	0	0	1	1
12.5	16	31	0	0	0	1	0
12.5	15	30	0	0	0	1	2
3.25	8	9	1	0	1	1	2
13	14	23	5	0	0	1	2
4.5	14	2	5	0	1	0	2
9.68	13	16	16	0	1	0	0
5	12	7	3	0	1	0	0
4.68	12	3	0	0	1	0	0
4.27	16	22	4	1	1	1	1
6.15	12	15	6	1	1	1	1
3.51	4	39	15	0	0	1	5
3	14	3	3	0	1	1	0
6.25	12	11	0	0	1	0	0
7.81	12	3	0	0	1	1	0

wage	educ	exper	tenure	nonwhite	female	married	numdep
10	12	20	5	1	1	1	4
4.5	14	16	0	0	1	1	1
4	11	45	12	0	1	1	0
6.38	13	11	4	0	1	0	0
13.7	15	20	13	0	0	1	2
1.67	10	1	0	0	0	0	1
2.93	12	36	2	0	1	1	1
3.65	14	9	2	0	0	0	0
2.9	12	15	1	0	1	1	2
1.63	12	18	0	0	1	0	2
8.6	16	3	2	0	1	0	0
5	12	15	5	0	0	1	1
6	12	7	7	0	0	1	0
2.5	12	2	0	0	0	0	2
3.25	15	3	0	0	0	0	1
3.4	16	1	1	0	1	0	1
10	8	13	0	1	0	0	0
21.63	18	8	8	0	1	0	0
4.38	16	7	0	0	0	0	0
11.71	13	40	20	0	1	0	0
12.39	14	42	5	0	0	0	0
6.25	10	36	8	0	0	0	0
3.71	10	13	0	1	1	0	4
7.78	14	9	3	0	0	0	0
19.98	14	26	23	0	0	1	2
6.25	16	7	4	1	1	1	3
10	12	25	3	0	0	1	3
5.71	16	10	5	0	0	1	1
2	12	3	2	0	1	0	4
5.71	16	3	0	0	1	0	0
13.08	17	17	2	1	0	1	3
4.91	12	17	8	0	0	1	2
2.91	12	20	34	0	1	1	2
3.75	12	7	0	0	1	1	0
11.9	13	24	19	0	0	1	2
4	12	28	0	0	1	1	1
3.1	12	2	1	0	1	0	1
8.45	12	19	13	0	0	1	4
7.14	18	13	0	0	0	1	2
4.5	9	22	5	0	0	1	4
4.65	16	3	1	0	0	0	0
2.9	10	4	0	0	1	0	1
6.67	12	7	5	0	0	0	0
3.5	12	6	2	1	1	0	1

wage	educ	exper	tenure	nonwhite	female	married	numdep
3.26	12	13	3	0	1	1	1
3.25	12	14	0	0	1	1	1
8	12	14	4	0	1	0	1
9.85	8	40	24	1	0	1	2
7.5	12	11	7	0	0	1	1
5.91	12	14	6	0	0	1	2
11.76	14	40	39	0	0	1	0
3	12	1	0	0	1	0	2
4.81	12	2	0	0	1	0	0
6.5	12	4	1	0	1	0	0
4	9	19	1	0	1	1	1
3.5	13	1	0	0	0	0	1
13.16	12	34	22	0	0	1	0
4.25	14	5	2	0	0	1	2
3.5	12	3	0	0	0	0	0

Display limited to first 100 rows and first 8 columns; complete data remain in the CSV/workbook.

Interpretation

This table contains 526 rows and 25 columns.

wage: 526 non-missing numeric entries; minimum 0.53, maximum 24.98, mean 5.896.

educ: 526 non-missing numeric entries; minimum 0, maximum 18, mean 12.56.

exper: 526 non-missing numeric entries; minimum 1, maximum 51, mean 17.02.

tenure: 526 non-missing numeric entries; minimum 0, maximum 44, mean 5.105.

nonwhite: 526 non-missing numeric entries; minimum 0, maximum 1, mean 0.1027.

female: 526 non-missing numeric entries; minimum 0, maximum 1, mean 0.4791.

married: 526 non-missing numeric entries; minimum 0, maximum 1, mean 0.6084.

numdep: 526 non-missing numeric entries; minimum 0, maximum 6, mean 1.044.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 68. Wage Profile

Source: /content/outputs_part1/wage_profile.csv

Unnamed: 0	wage
count	526
min	0.53
1st percentile	1.69
median	4.65
99th percentile	19.995
max	24.98
values <= 0	0
values missing	0
values above Tukey upper fence (Q3 + 1.5 IQR)	36

Interpretation

This table contains 9 rows and 2 columns.

wage: 9 non-missing numeric entries; minimum 0, maximum 526, mean 68.2.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 69. Individual T Tests

Source: /content/outputs_part2/tables/individual_t_tests.csv

Variable	Model	Coefficient	Standard error	t-statistic	p-value	95% CI lower	95% CI upper
const	Simple OLS	-0.9049	0.685	-1.321	0.1871	-2.2505	0.4408
educ	Simple OLS	0.5414	0.0532	10.1667	2.783e-22	0.4368	0.646
Intercept	Multiple OLS	-1.6022	0.7311	-2.1916	0.0289	-3.0384	-0.166
educ	Multiple OLS	0.5551	0.0501	11.0897	8.574e-26	0.4568	0.6534
exper	Multiple OLS	0.0187	0.012	1.557	0.1201	-0.0049	0.0424
tenure	Multiple OLS	0.1388	0.0212	6.5618	1.290e-10	0.0973	0.1804
female	Multiple OLS	-1.7424	0.2668	-6.5303	1.566e-10	-2.2666	-1.2182
married	Multiple OLS	0.5566	0.2867	1.941	0.0528	-0.0068	1.1199
nonwhite	Multiple OLS	-0.0658	0.4266	-0.1543	0.8775	-0.9038	0.7722

Display limited to first 100 rows and first 8 columns; complete data remain in the CSV/workbook.

Interpretation

This table contains 9 rows and 9 columns.

Estimated coefficient for const is -0.9049 (p-value 0.1871); not statistically significant at the 5% level under the reported model.

Estimated coefficient for educ is 0.5414 (p-value 2.783e-22); statistically significant at the 5% level under the reported model.

Estimated coefficient for Intercept is -1.6022 (p-value 0.02885); statistically significant at the 5% level under the reported model.

Estimated coefficient for educ is 0.5551 (p-value 8.574e-26); statistically significant at the 5% level under the reported model.

Estimated coefficient for exper is 0.0187 (p-value 0.1201); not statistically significant at the 5% level under the reported model.

Estimated coefficient for tenure is 0.1388 (p-value 1.29e-10); statistically significant at the 5% level under the reported model.

Estimated coefficient for female is -1.7424 (p-value 1.566e-10); statistically significant at the 5% level under the reported model.

Estimated coefficient for married is 0.5566 (p-value 0.0528); not statistically significant at the 5% level under the reported model.

Estimated coefficient for nonwhite is -0.0658 (p-value 0.8775); not statistically significant at the 5% level under the reported model.

Table 70. Joint F Test Experience Tenure

Source: /content/outputs_part2/tables/joint_f_test_experience_tenure.csv

Test	Null hypothesis	F-statistic	Numerator df	Denominator df	p-value	Reject H0 at 5%
Joint significance of experience and tenure	beta_exper = 0 and beta_tenure = 0	35.2061	2	519	4.594e-15	True

Interpretation

This table contains 1 rows and 7 columns.

Joint significance of experience and tenure: p-value = 4.594e-15; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Table 71. Model Sample Descriptive Statistics

Source: /content/outputs_part2/tables/model_sample_descriptive_statistics.csv

Unnamed: 0	count	mean	median	std	min	max
wage	526	5.8961	4.65	3.6931	0.53	24.98
educ	526	12.5627	12	2.769	0	18
exper	526	17.0171	13.5	13.5722	1	51
tenure	526	5.1046	2	7.2245	0	44

Unnamed: 0	count	mean	median	std	min	max
female	526	0.4791	0	0.5	0	1
married	526	0.6084	1	0.4886	0	1
nonwhite	526	0.1027	0	0.3038	0	1

Interpretation

This table contains 7 rows and 7 columns.

count: 7 non-missing numeric entries; minimum 526, maximum 526, mean 526.

mean: 7 non-missing numeric entries; minimum 0.1027, maximum 17.02, mean 5.967.

median: 7 non-missing numeric entries; minimum 0, maximum 13.5, mean 4.736.

std: 7 non-missing numeric entries; minimum 0.3038, maximum 13.57, mean 4.079.

min: 7 non-missing numeric entries; minimum 0, maximum 1, mean 0.2186.

max: 7 non-missing numeric entries; minimum 1, maximum 51, mean 20.14.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 72. Multiple Model Interpretation

Source: /content/outputs_part2/tables/multiple_model_interpretation.csv

Variable	Estimated coefficient	p-value	Statistical significance at 5%	Interpretation
educ	0.5551	8.574e-26	Statistically significant	Holding the other included regressors constant, a one-unit increase in educ is associated with an estimated 0.5551-unit change in hourly wage. This is an association, not necessarily a causal effect.
exper	0.0187	0.1201	Not statistically significant	Holding the other included regressors constant, a one-unit increase in exper is associated with an estimated 0.0187-unit change in hourly wage. This is an association, not necessarily a causal effect.
tenure	0.1388	1.290e-10	Statistically significant	Holding the other included regressors constant, a one-unit increase in tenure is associated with an estimated 0.1388-unit change in hourly wage. This is an association, not necessarily a causal effect.
female	-1.7424	1.566e-10	Statistically significant	Holding the other included regressors constant, a one-unit increase in female is associated with an estimated -1.7424-unit change in hourly wage. This is an association, not necessarily a causal effect.

Variable	Estimated coefficient	p-value	Statistical significance at 5%	Interpretation
married	0.5566	0.0528	Not statistically significant	Holding the other included regressors constant, a one-unit increase in married is associated with an estimated 0.5566-unit change in hourly wage. This is an association, not necessarily a causal effect.
nonwhite	-0.0658	0.8775	Not statistically significant	Holding the other included regressors constant, a one-unit increase in nonwhite is associated with an estimated -0.0658-unit change in hourly wage. This is an association, not necessarily a causal effect.

Interpretation

This table contains 6 rows and 5 columns.

educ: p-value = 8.574e-26; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

exper: p-value = 0.1201; do not reject the null hypothesis at the 5% level, conditional on the test's assumptions.

tenure: p-value = 1.29e-10; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

female: p-value = 1.566e-10; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

married: p-value = 0.0528; do not reject the null hypothesis at the 5% level, conditional on the test's assumptions.

nonwhite: p-value = 0.8775; do not reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Table 73. Ols Coefficient Comparison

Source: /content/outputs_part2/tables/ols_coefficient_comparison.csv

Model	Variable	Coefficient	Standard error	t-statistic	p-value	95% CI lower	95% CI upper
Simple OLS	const	-0.9049	0.685	-1.321	0.1871	-2.2505	0.4408
Simple OLS	educ	0.5414	0.0532	10.1667	2.783e-22	0.4368	0.646
Multiple OLS	Intercept	-1.6022	0.7311	-2.1916	0.0289	-3.0384	-0.166
Multiple OLS	educ	0.5551	0.0501	11.0897	8.574e-26	0.4568	0.6534
Multiple OLS	exper	0.0187	0.012	1.557	0.1201	-0.0049	0.0424
Multiple OLS	tenure	0.1388	0.0212	6.5618	1.290e-10	0.0973	0.1804
Multiple OLS	female	-1.7424	0.2668	-6.5303	1.566e-10	-2.2666	-1.2182
Multiple OLS	married	0.5566	0.2867	1.941	0.0528	-0.0068	1.1199
Multiple OLS	nonwhite	-0.0658	0.4266	-0.1543	0.8775	-0.9038	0.7722

Interpretation

This table contains 9 rows and 8 columns.

Estimated coefficient for const is -0.9049 (p-value 0.1871); not statistically significant at the 5% level under the reported model.

Estimated coefficient for educ is 0.5414 (p-value 2.783e-22); statistically significant at the 5% level under the reported model.

Estimated coefficient for Intercept is -1.6022 (p-value 0.02885); statistically significant at the 5% level under the reported model.

Estimated coefficient for educ is 0.5551 (p-value 8.574e-26); statistically significant at the 5% level under the reported model.

Estimated coefficient for exper is 0.0187 (p-value 0.1201); not statistically significant at the 5% level under the reported model.

Estimated coefficient for tenure is 0.1388 (p-value 1.29e-10); statistically significant at the 5% level under the reported model.

Estimated coefficient for female is -1.7424 (p-value 1.566e-10); statistically significant at the 5% level under the reported model.

Estimated coefficient for married is 0.5566 (p-value 0.0528); not statistically significant at the 5% level under the reported model.

Estimated coefficient for nonwhite is -0.0658 (p-value 0.8775); not statistically significant at the 5% level under the reported model.

Table 74. Ols Model Fit Comparison

Source: /content/outputs_part2/tables/ols_model_fit_comparison.csv

Model	Dependent variable	Observations	R-squared	Adjusted R-squared	Residual standard error	F-statistic	F-test p-value
Simple OLS	wage	526	0.1648	0.1632	3.3784	103.3627	2.783e-22
Multiple OLS	wage	526	0.3682	0.3609	2.9524	50.4142	8.269e-49

Interpretation

This table contains 2 rows and 8 columns.

Simple OLS: F-test p-value = 2.783e-22; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Multiple OLS: F-test p-value = 8.269e-49; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Table 75. Overall F Tests

Source: /content/outputs_part2/tables/overall_f_tests.csv

Model	Null hypothesis	F-statistic	Numerator df	Denominator df	p-value	Reject H0 at 5%
Simple OLS	All slope coefficients equal zero	103.3627	1	524	2.783e-22	True
Multiple OLS	All slope coefficients equal zero	50.4142	6	519	8.269e-49	True

Interpretation

This table contains 2 rows and 7 columns.

Simple OLS: p-value = 2.783e-22; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Multiple OLS: p-value = 8.269e-49; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Table 76. Sample Summary

Source: /content/outputs_part2/tables/sample_summary.csv

Unnamed: 0	Initial observations	Complete observations	Observations excluded for missing/non-finite values
wage1 model sample	526	526	0

Interpretation

This table contains 1 rows and 4 columns.

Initial observations: 1 non-missing numeric entries; minimum 526, maximum 526, mean 526.

Complete observations: 1 non-missing numeric entries; minimum 526, maximum 526, mean 526.

Observations excluded for missing/non-finite values: 1 non-missing numeric entries; minimum 0, maximum 0, mean 0.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 77. Education Indicator Interaction

Source: /content/outputs_part3/tables/education_indicator_interaction.csv

model	term	coefficient	std_error	t_statistic	p_value	ci_lower_95	ci_upper_95
Log wage with education interaction	Intercept	0.4553	0.1235	3.6877	2.502e-04	0.2127	0.6978

model	term	coefficient	std_error	t_statistic	p_value	ci_lower_95	ci_upper_95
Log wage with education interaction	educ	0.0866	0.0088	9.8706	3.587e-21	0.0694	0.1038
Log wage with education interaction	exper	0.0031	0.0017	1.8678	0.0623	-1.630e-04	0.0065
Log wage with education interaction	tenure	0.0169	0.003	5.7143	1.861e-08	0.0111	0.0228
Log wage with education interaction	female	-0.199	0.1731	-1.1496	0.2508	-0.5391	0.1411
Log wage with education interaction	married	0.1256	0.0401	3.129	0.0019	0.0467	0.2044
Log wage with education interaction	nonwhite	-5.823e-04	0.0598	-0.0097	0.9922	-0.1181	0.1169
Log wage with education interaction	educ:female	-0.0069	0.0135	-0.5121	0.6088	-0.0334	0.0196

Interpretation

This table contains 8 rows and 8 columns.

Estimated coefficient for Intercept is 0.4553 (p-value 0.0002502); statistically significant at the 5% level under the reported model.

Estimated coefficient for educ is 0.0866 (p-value 3.587e-21); statistically significant at the 5% level under the reported model.

Estimated coefficient for exper is 0.0031 (p-value 0.06235); not statistically significant at the 5% level under the reported model.

Estimated coefficient for tenure is 0.0169 (p-value 1.861e-08); statistically significant at the 5% level under the reported model.

Estimated coefficient for female is -0.1990 (p-value 0.2508); not statistically significant at the 5% level under the reported model.

Estimated coefficient for married is 0.1256 (p-value 0.001853); statistically significant at the 5% level under the reported model.

Estimated coefficient for nonwhite is -0.0006 (p-value 0.9922); not statistically significant at the 5% level under the reported model.

Estimated coefficient for educ:female is -0.0069 (p-value 0.6088); not statistically significant at the 5% level under the reported model.

Table 78. Functional Form Coefficients

Source: /content/outputs_part3/tables/functional_form_coefficients.csv

model	term	coefficient	std_error	t_statistic	p_value	ci_lower_95	ci_upper_95
Linear wage	Intercept	-1.6022	0.7311	-2.1916	0.0289	-3.0384	-0.166
Linear wage	educ	0.5551	0.0501	11.0897	8.574e-26	0.4568	0.6534
Linear wage	exper	0.0187	0.012	1.557	0.1201	-0.0049	0.0424
Linear wage	tenure	0.1388	0.0212	6.5618	1.290e-10	0.0973	0.1804
Linear wage	female	-1.7424	0.2668	-6.5303	1.566e-10	-2.2666	-1.2182
Linear wage	married	0.5566	0.2867	1.941	0.0528	-0.0068	1.1199
Linear wage	nonwhite	-0.0658	0.4266	-0.1543	0.8775	-0.9038	0.7722
Log wage	Intercept	0.4907	0.1022	4.7996	2.082e-06	0.2898	0.6915
Log wage	educ	0.0839	0.007	11.9839	2.198e-29	0.0701	0.0976
Log wage	exper	0.0031	0.0017	1.8614	0.0632	-1.736e-04	0.0064
Log wage	tenure	0.0169	0.003	5.7017	1.993e-08	0.0111	0.0227
Log wage	female	-0.2856	0.0373	-7.6537	9.590e-14	-0.3589	-0.2123
Log wage	married	0.1256	0.0401	3.1333	0.0018	0.0469	0.2044
Log wage	nonwhite	-0.0025	0.0597	-0.0424	0.9662	-0.1197	0.1147

Interpretation

This table contains 14 rows and 8 columns.

Estimated coefficient for Intercept is -1.6022 (p-value 0.02885); statistically significant at the 5% level under the reported model.

Estimated coefficient for educ is 0.5551 (p-value 8.574e-26); statistically significant at the 5% level under the reported model.

Estimated coefficient for exper is 0.0187 (p-value 0.1201); not statistically significant at the 5% level under the reported model.

Estimated coefficient for tenure is 0.1388 (p-value 1.29e-10); statistically significant at the 5% level under the reported model.

Estimated coefficient for female is -1.7424 (p-value 1.566e-10); statistically significant at the 5% level under the reported model.

Estimated coefficient for married is 0.5566 (p-value 0.0528); not statistically significant at the 5% level under the reported model.

Estimated coefficient for nonwhite is -0.0658 (p-value 0.8775); not statistically significant at the 5% level under the reported model.

Estimated coefficient for Intercept is 0.4907 (p-value 2.082e-06); statistically significant at the 5% level under the reported model.

Estimated coefficient for educ is 0.0839 (p-value 2.198e-29); statistically significant at the 5% level under the reported model.

Estimated coefficient for exper is 0.0031 (p-value 0.06325); not statistically significant at the 5% level under the reported model.

Estimated coefficient for tenure is 0.0169 (p-value 1.993e-08); statistically significant at the 5% level under the reported model.

Estimated coefficient for female is -0.2856 (p-value 9.59e-14); statistically significant at the 5% level under the reported model.

Estimated coefficient for married is 0.1256 (p-value 0.001826); statistically significant at the 5% level under the reported model.

Estimated coefficient for nonwhite is -0.0025 (p-value 0.9662); not statistically significant at the 5% level under the reported model.

Table 79. Functional Form Model Summary

Source: /content/outputs_part3/tables/functional_form_model_summary.csv

model	dependent_variable	n	r_squared	adjusted_r_squared	aic
Linear wage	wage	526	0.3682	0.3609	2638.5762
Log wage	log_wage	526	0.4036	0.3967	569.0077

Interpretation

This table contains 2 rows and 6 columns.

n: 2 non-missing numeric entries; minimum 526, maximum 526, mean 526.

r_squared: 2 non-missing numeric entries; minimum 0.3682, maximum 0.4036, mean 0.3859.

adjusted_r_squared: 2 non-missing numeric entries; minimum 0.3609, maximum 0.3967, mean 0.3788.

aic: 2 non-missing numeric entries; minimum 569, maximum 2639, mean 1604.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 80. Joint Dummy F Test

Source: /content/outputs_part3/tables/joint_dummy_f_test.csv

test	null	f_statistic	df_num	df_denom	p_value	reject_at_5pct
Joint F-test: worker-characteristic indicators	female = married = nonwhite = 0	25.411	3	519	2.366e-15	True

Interpretation

This table contains 1 rows and 7 columns.

Joint F-test: worker-characteristic indicators: p_value = 2.366e-15; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Table 81. Log Wage Coefficient Interpretation

Source: /content/outputs_part3/tables/log_wage_coefficient_interpretation.csv

variable	coefficient	implied_percent_difference	indicator
educ	0.0839	8.7501	False
exper	0.0031	0.3139	False
tenure	0.0169	1.7012	False
female	-0.2856	-24.8413	True
married	0.1256	13.387	True
nonwhite	-0.0025	-0.2524	True

Interpretation

This table contains 6 rows and 4 columns.

Estimated coefficient for educ is 0.0839.

Estimated coefficient for exper is 0.0031.

Estimated coefficient for tenure is 0.0169.

Estimated coefficient for female is -0.2856.

Estimated coefficient for married is 0.1256.

Estimated coefficient for nonwhite is -0.0025.

Table 82. Mwd Functional Form Test

Source: /content/outputs_part3/tables/mwd_functional_form_test.csv

status	reason
Not computed	Nonpositive or nonfinite fitted wages prevent log(fitted wage).

Interpretation

This table contains 1 rows and 2 columns.

This is a categorical/text summary table; interpret its entries using the row and column labels.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 83. Conventional Vs Robust Standard Errors

Source: /content/outputs_part4/tables/conventional_vs_robust_standard_errors.csv

Variable	Coefficient	Conventional standard error	Robust standard error (HC1)	Conventional p-value	Robust p-value (HC1)
Intercept	-1.6022	0.7311	0.8227	0.0289	0.052
educ	0.5551	0.0501	0.0626	8.574e-26	1.158e-17
exper	0.0187	0.012	0.0101	0.1201	0.065
tenure	0.1388	0.0212	0.028	1.290e-10	9.613e-07
female	-1.7424	0.2668	0.2504	1.566e-10	1.038e-11
married	0.5566	0.2867	0.2593	0.0528	0.0323
nonwhite	-0.0658	0.4266	0.3888	0.8775	0.8657

Interpretation

This table contains 7 rows and 6 columns.

Estimated coefficient for Intercept is -1.6022.

Estimated coefficient for educ is 0.5551.

Estimated coefficient for exper is 0.0187.

Estimated coefficient for tenure is 0.1388.

Estimated coefficient for female is -1.7424.

Estimated coefficient for married is 0.5566.

Estimated coefficient for nonwhite is -0.0658.

Table 84. Diagnostic Summary

Source: /content/outputs_part4/tables/diagnostic_summary.csv

Model	Observations	R-squared	Adjusted R-squared	Breusch-Pagan-Godfrey p-value	White test p-value	Jarque-Bera p-value
Multiple OLS: wage ~ educ + exper + tenure + female + married + nonwhite	526	0.3682	0.3609	6.701e-08	5.624e-07	8.109e-159

Interpretation

This table contains 1 rows and 7 columns.

Multiple OLS: wage ~ educ + exper + tenure + female + married + nonwhite: Breusch-Pagan-Godfrey p-value = 6.701e-08; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Multiple OLS: wage ~ educ + exper + tenure + female + married + nonwhite: White test p-value = 5.624e-07; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Multiple OLS: wage ~ educ + exper + tenure + female + married + nonwhite: Jarque-Bera p-value = 8.109e-159; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Table 85. Heteroskedasticity Tests

Source: /content/outputs_part4/tables/heteroskedasticity_tests.csv

Test	LM statistic	LM p-value	F statistic	F p-value	Reject homoskedasticity at 5%
Breusch-Pagan-Godfrey	44.2148	6.701e-08	7.9384	3.329e-08	True
White	73.8515	5.624e-07	3.4096	1.552e-07	True

Interpretation

This table contains 2 rows and 6 columns.

Breusch-Pagan-Godfrey: LM p-value = 6.701e-08; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Breusch-Pagan-Godfrey: F p-value = 3.329e-08; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

White: LM p-value = 5.624e-07; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

White: F p-value = 1.552e-07; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Table 86. Jarque Bera Normality Test

Source: /content/outputs_part4/tables/jarque_bera_normality_test.csv

Test	Statistic	p-value	Reject normality at 5%	Number of residuals
Jarque-Bera	728.0361	8.109e-159	True	526

Interpretation

This table contains 1 rows and 5 columns.

Jarque-Bera: p-value = 8.109e-159; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Table 87. Predictor Correlation Matrix

Source: /content/outputs_part4/tables/predictor_correlation_matrix.csv

Unnamed: 0	educ	exper	tenure	female	married	nonwhite
educ	1	-0.2995	-0.0562	-0.085	0.0689	-0.0847

Unnamed: 0	educ	exper	tenure	female	married	nonwhite
exper	-0.2995	1	0.4993	-0.0416	0.317	0.0144
tenure	-0.0562	0.4993	1	-0.1979	0.2399	0.0116
female	-0.085	-0.0416	-0.1979	1	-0.1661	-0.0109
married	0.0689	0.317	0.2399	-0.1661	1	-0.0623
nonwhite	-0.0847	0.0144	0.0116	-0.0109	-0.0623	1

Interpretation

This table contains 6 rows and 7 columns.

educ: 6 non-missing numeric entries; minimum -0.2995, maximum 1, mean 0.09058.

exper: 6 non-missing numeric entries; minimum -0.2995, maximum 1, mean 0.2482.

tenure: 6 non-missing numeric entries; minimum -0.1979, maximum 1, mean 0.2494.

female: 6 non-missing numeric entries; minimum -0.1979, maximum 1, mean 0.08306.

married: 6 non-missing numeric entries; minimum -0.1661, maximum 1, mean 0.2329.

nonwhite: 6 non-missing numeric entries; minimum -0.08465, maximum 1, mean 0.1447.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 88. Variance Inflation Factors

Source: /content/outputs_part4/tables/variance_inflation_factors.csv

Variable	VIF
educ	1.1571
exper	1.6084
tenure	1.4072
female	1.0722
married	1.1822
nonwhite	1.0115

Interpretation

This table contains 6 rows and 2 columns.

VIF: 6 non-missing numeric entries; minimum 1.012, maximum 1.608, mean 1.24.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 89. Autocorrelation Tests

Source: /content/outputs_part5/tables/autocorrelation_tests.csv

Test	Statistic	p-value	Lag order	Interpretation
Durbin-Watson	0.8015		1	The Durbin-Watson statistic is descriptive; use its appropriate critical-value bounds or a suitable formal procedure for inference.
Breusch-Godfrey	26.0731	3.059e-05	4	Tests the null of no residual autocorrelation up to the specified lag order.

Interpretation

This table contains 2 rows and 5 columns.

Durbin-Watson: p-value = nan; do not reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Breusch-Godfrey: p-value = 3.059e-05; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Table 90. Breusch Godfrey F Test

Source: /content/outputs_part5/tables/breusch_godfrey_f_test.csv

Test	F-statistic	p-value	Lag order
Breusch-Godfrey F form	10.8903	1.989e-06	4

Interpretation

This table contains 1 rows and 4 columns.

Breusch-Godfrey F form: p-value = 1.989e-06; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Table 91. Joint F Test Selected Regressors

Source: /content/outputs_part5/tables/joint_f_test_selected_regressors.csv

Test	Restrictions	F-statistic	Numerator df	Denominator df	p-value	Reject null at 5%
Joint F-test for selected regressors	married = 0 and nonwhite = 0	1.9208	2	519	0.1475	False

Interpretation

This table contains 1 rows and 7 columns.

Joint F-test for selected regressors: p-value = 0.1475; do not reject the null hypothesis at the 5% level, conditional on the test's assumptions.

Table 92. Phillips Time Series Data Notes

Source: /content/outputs_part5/tables/phillips_time_series_data_notes.csv

Observations used	Dependent variable	Explanatory variable	Ordering	Time index verified from metadata	Interpretation note
56	inf	unem	Original dataset row order	False	Autocorrelation tests require meaningful temporal ordering. Confirm the dataset's documented time index before interpreting these results as time-series evidence.

Interpretation

This table contains 1 rows and 6 columns.

Observations used: 1 non-missing numeric entries; minimum 56, maximum 56, mean 56.

Dependent variable: 1 non-missing numeric entries; minimum inf, maximum inf, mean inf.

Interpretation is descriptive of the exported values; statistical or causal claims should be based on the relevant model specification and assumptions.

Table 93. Ramsey Reset Test

Source: /content/outputs_part5/tables/ramsey_reset_test.csv

Test	Specification	F-statistic	p-value	Reject null at 5%
Ramsey RESET	Powers of fitted values through cubic	15.1683	3.968e-07	True

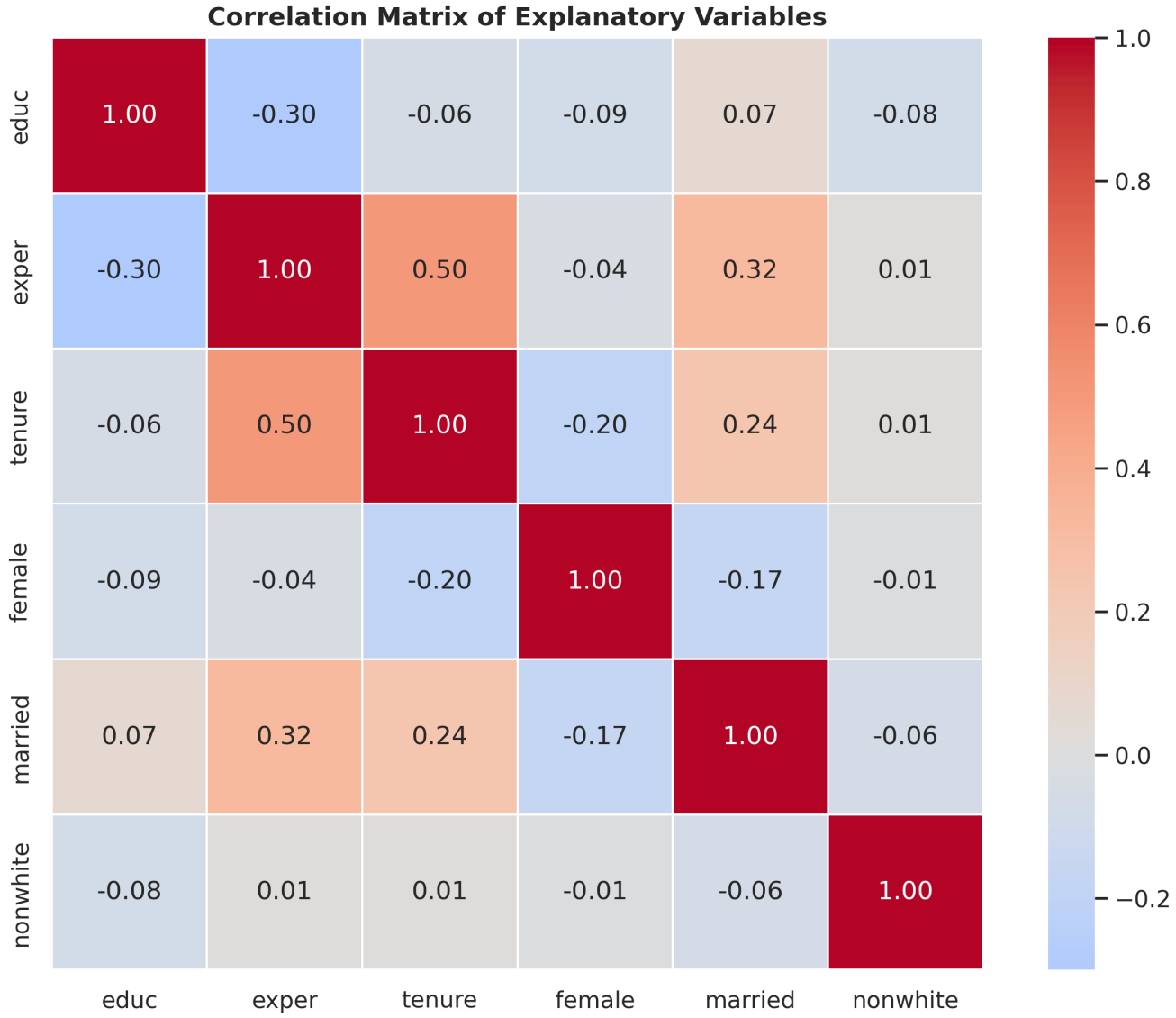
Interpretation

This table contains 1 rows and 5 columns.

Ramsey RESET: p-value = 3.968e-07; reject the null hypothesis at the 5% level, conditional on the test's assumptions.

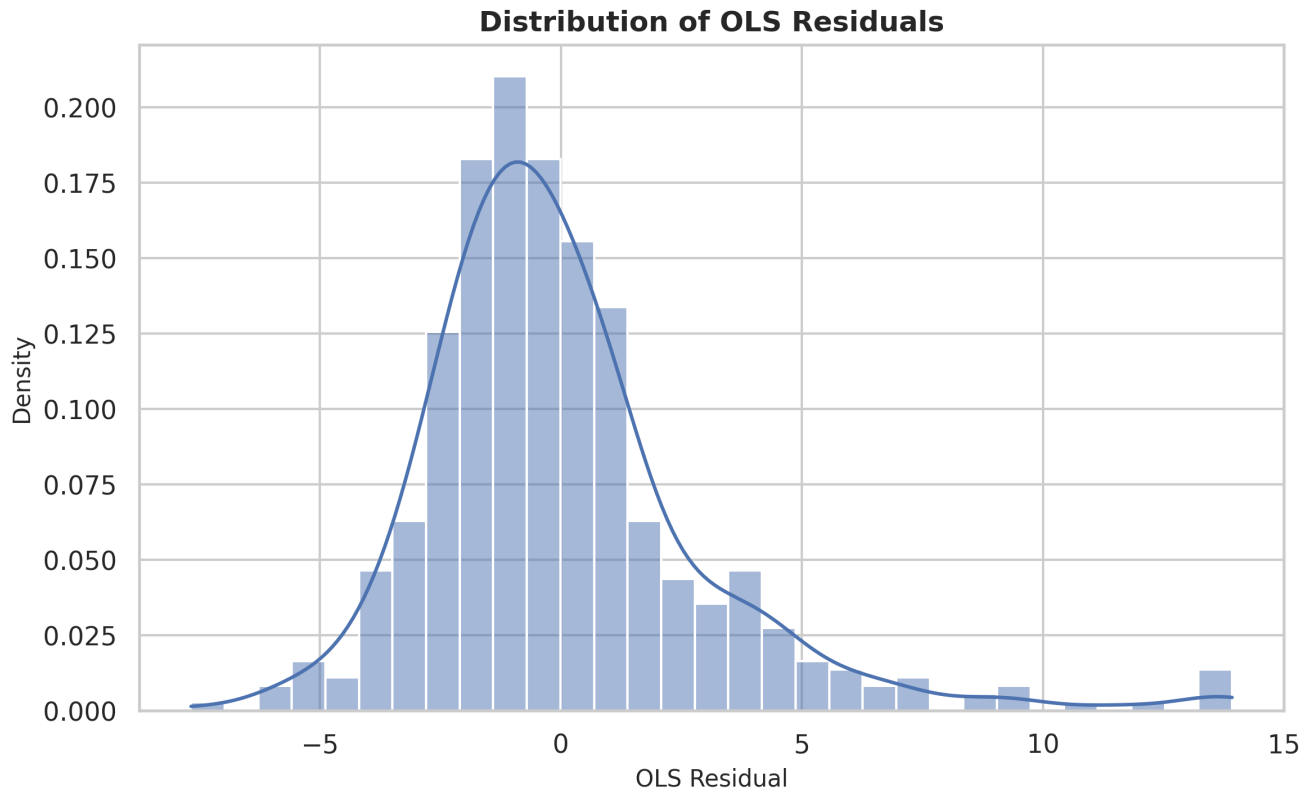
Figures

Figure 1. Predictor Correlation Heatmap



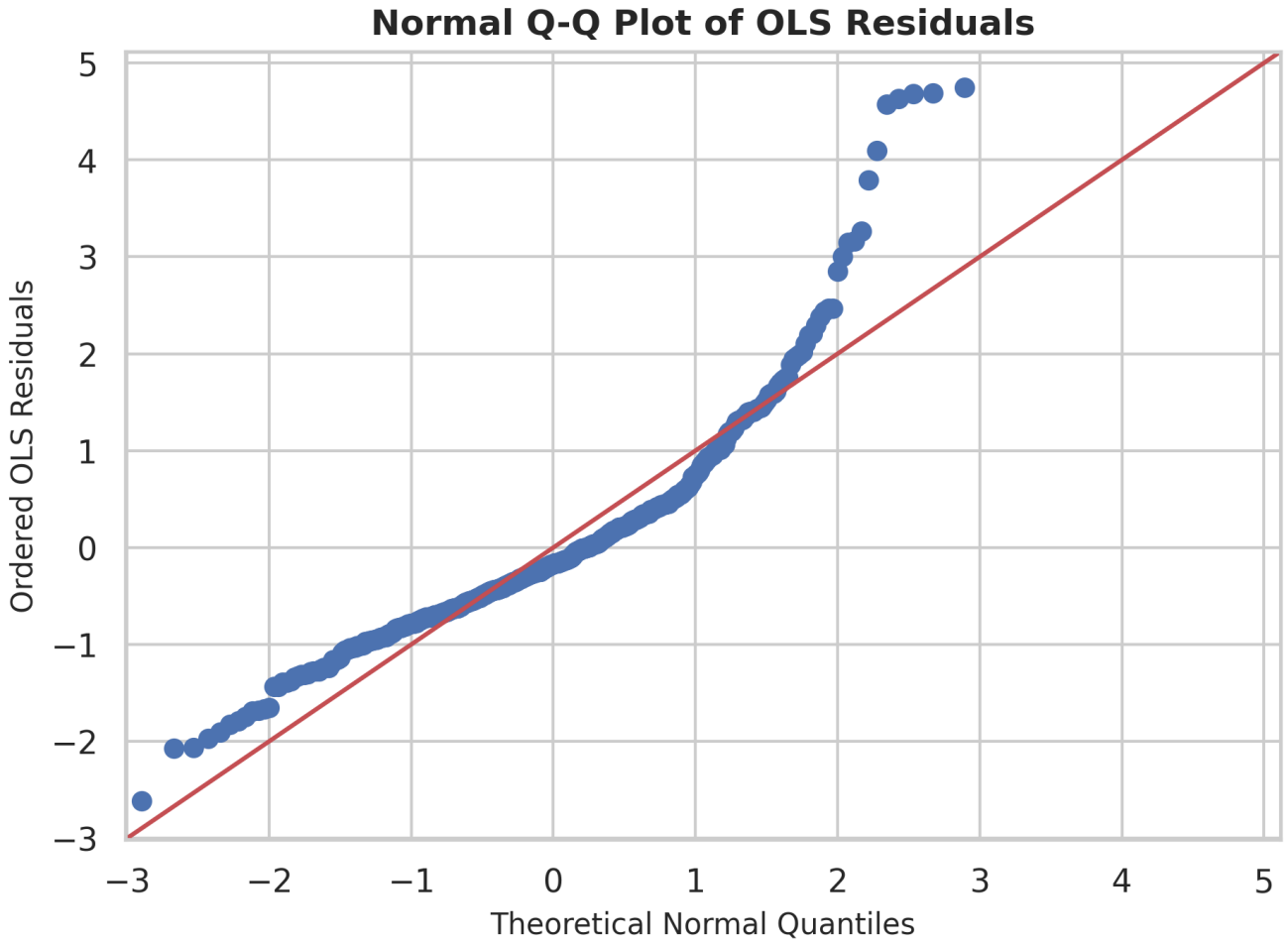
Interpretation: use the title and axes to identify the displayed pattern and compare it with the relevant descriptive or model results. A figure alone does not establish causality.

Figure 2. Residual Distribution



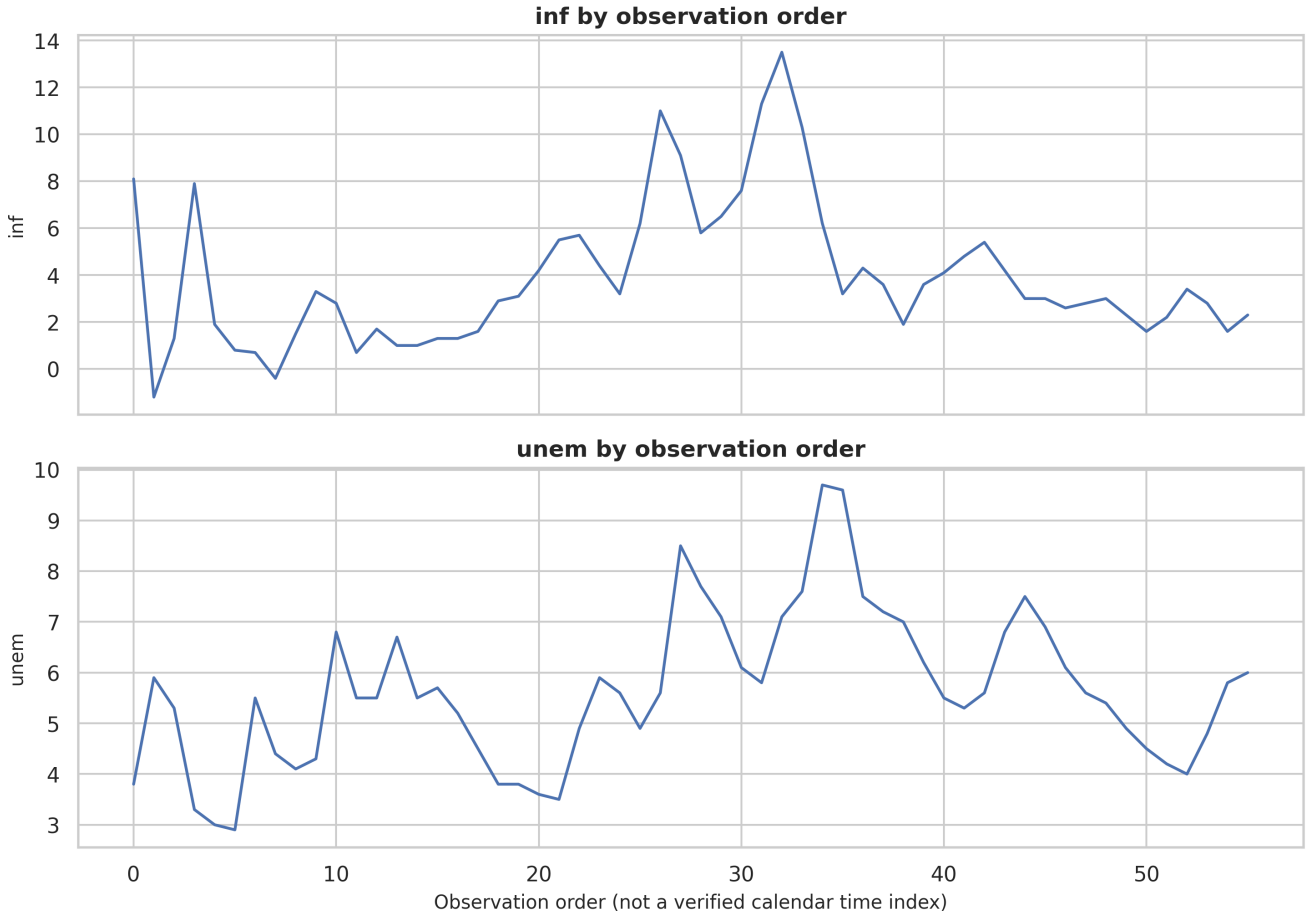
Interpretation: use the title and axes to identify the displayed pattern and compare it with the relevant descriptive or model results. A figure alone does not establish causality.

Figure 3. Residual Normal Qq Plot



Interpretation: use the title and axes to identify the displayed pattern and compare it with the relevant descriptive or model results. A figure alone does not establish causality.

Figure 4. Phillips Variables By Observation Order



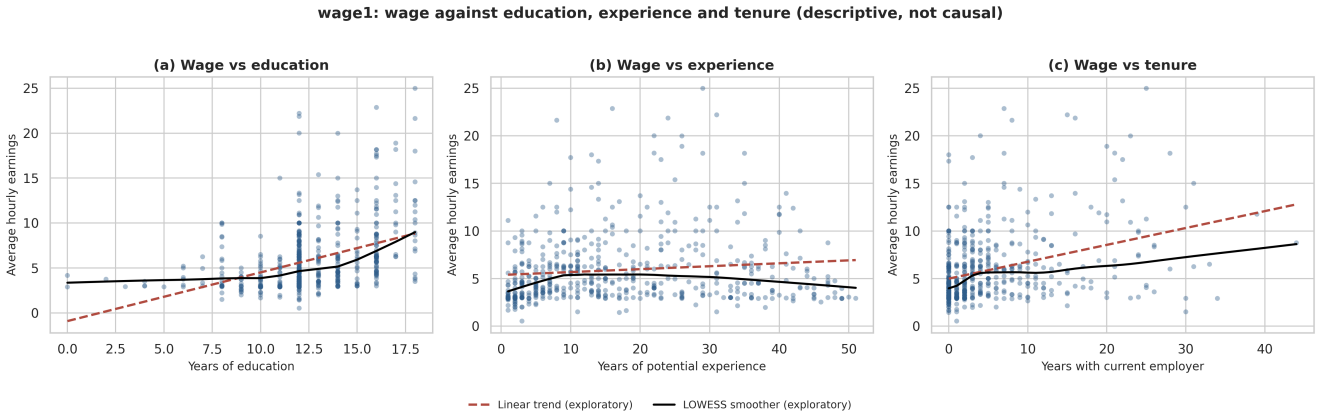
Interpretation: use the title and axes to identify the displayed pattern and compare it with the relevant descriptive or model results. A figure alone does not establish causality.

Figure 5. Fig1 Wage Distribution



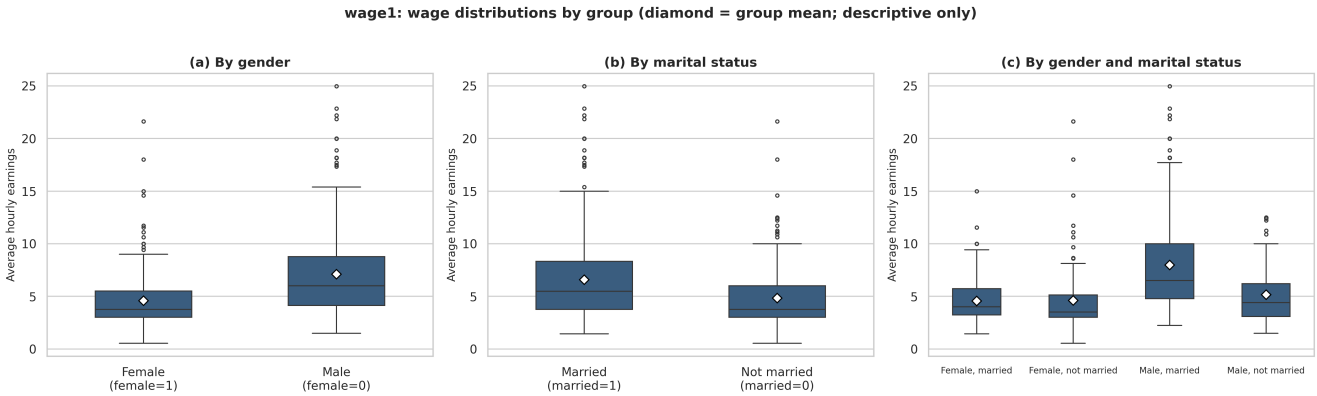
Interpretation: use the title and axes to identify the displayed pattern and compare it with the relevant descriptive or model results. A figure alone does not establish causality.

Figure 6. Fig2 Wage Scatterplots



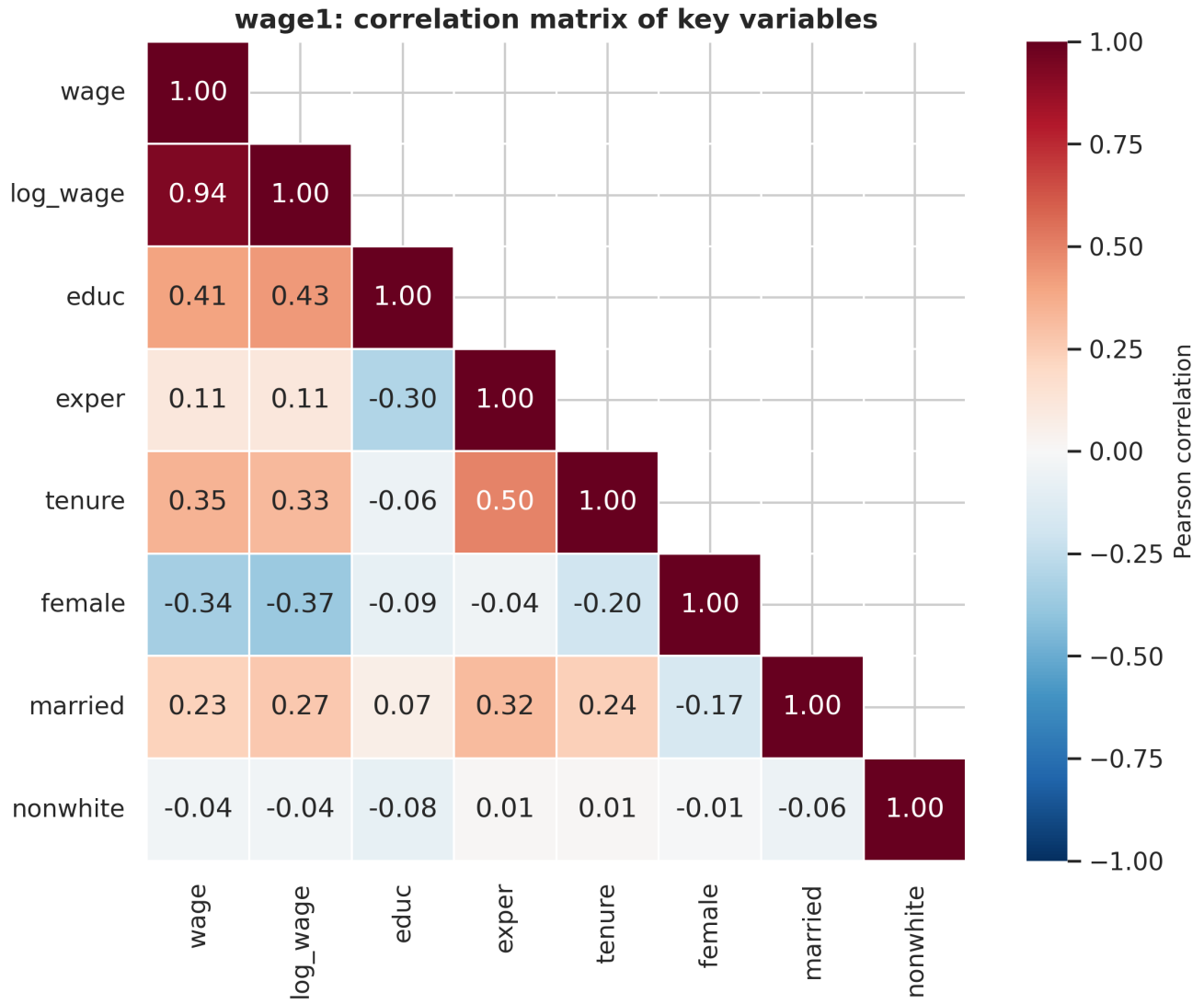
Interpretation: use the title and axes to identify the displayed pattern and compare it with the relevant descriptive or model results. A figure alone does not establish causality.

Figure 7. Fig3 Wage By Groups



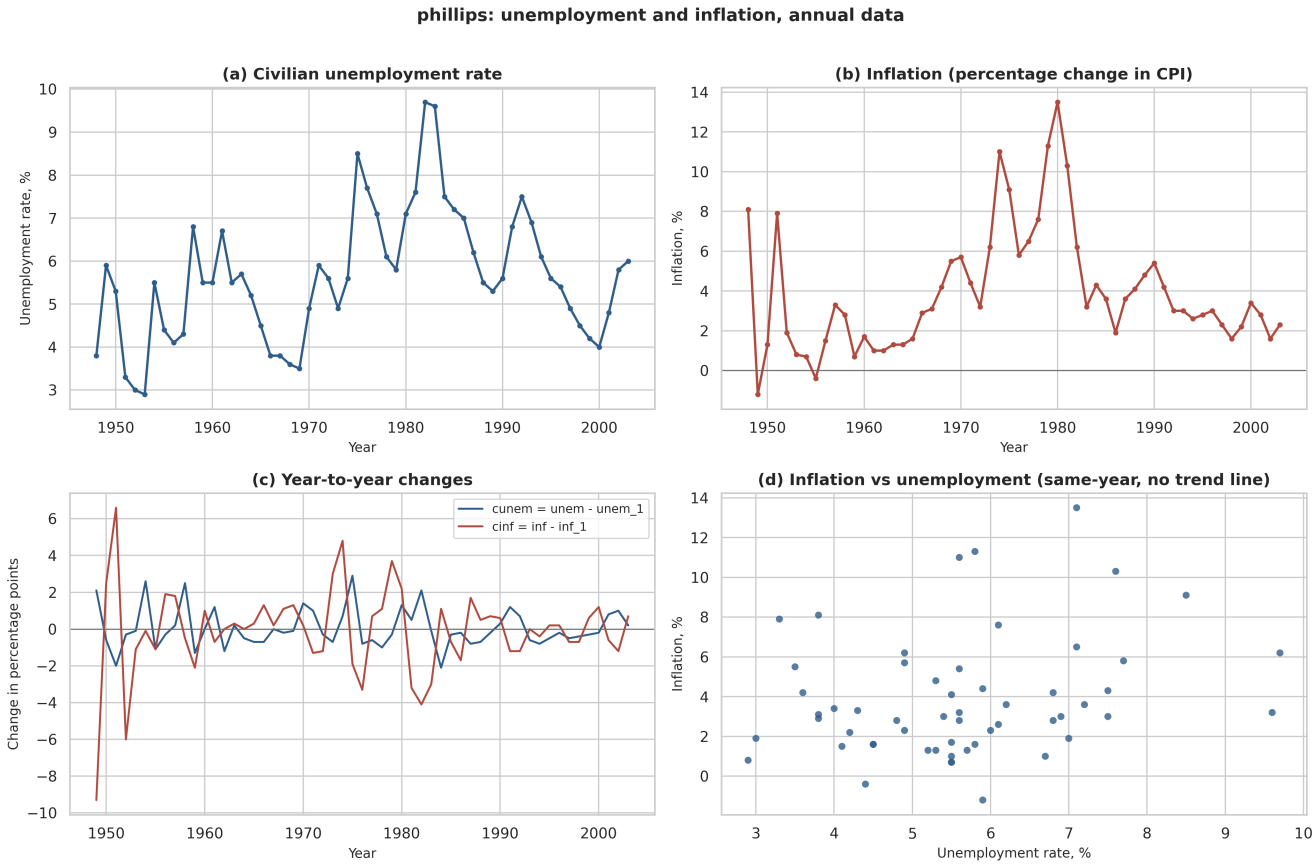
Interpretation: use the title and axes to identify the displayed pattern and compare it with the relevant descriptive or model results. A figure alone does not establish causality.

Figure 8. Fig4 Correlation Heatmap



Interpretation: use the title and axes to identify the displayed pattern and compare it with the relevant descriptive or model results. A figure alone does not establish causality.

Figure 9. Fig5 Phillips Time Plots



Interpretation: use the title and axes to identify the displayed pattern and compare it with the relevant descriptive or model results. A figure alone does not establish causality.

Source code appendix

The following appendix reproduces available Python source files. Code is included as text for reference; run the original .py scripts in the appropriate order in Colab.

Part 1 source code

Source file for Part 1 was not found. Upload/copy it into /content and rerun this generator.

Part 2 source code

Source file for Part 2 was not found. Upload/copy it into /content and rerun this generator.

Part 3 source code

Source file for Part 3 was not found. Upload/copy it into /content and rerun this generator.

Part 4 source code

Source file for Part 4 was not found. Upload/copy it into /content and rerun this generator.

Part 5 source code

Source file for Part 5 was not found. Upload/copy it into /content and rerun this generator.

Part 6 source code

Source file for Part 6 was not found. Upload/copy it into /content and rerun this generator.

Reproducibility and limitations

Only artifacts saved to disk can be included automatically. Results that were displayed but not exported cannot be recovered by this script. Automatically generated interpretations are deliberately cautious and should be checked against the original regression specification and assumptions before submission. Statistical significance is not equivalent to causality.